

#1 · Cardiovascular Emergencies

What are the five must-not-miss life threats in a patient with undifferentiated chest pain?

Quick Reference

Acute coronary syndrome, pulmonary embolism, aortic dissection, tension pneumothorax, and esophageal rupture (Boerhaave syndrome). Every chest pain workup should actively rule these out before settling on a benign cause. Missing one of these is the classic trap on the CCFP(EM) exam.

#2 · Cardiovascular Emergencies

Which patient groups commonly present with atypical or silent myocardial ischemia?

Quick Reference

Women, older adults, and people with diabetes often present with fatigue, nausea, epigastric discomfort, or dyspnea instead of classic crushing substernal pain. Diabetic autonomic neuropathy blunts pain signaling. A high index of suspicion is needed since a benign-looking presentation does not rule out ACS in these groups.

#3 · Cardiovascular Emergencies

What clinical clues point to aortic dissection rather than ACS in a patient with chest pain?

Quick Reference

Tearing or ripping pain that radiates to the back, a pulse deficit or a blood pressure difference of more than 20 mmHg between arms, and a widened mediastinum on chest x-ray all suggest dissection. Giving thrombolytics for a presumed MI in an undiagnosed dissection can be fatal, so these clues must be actively sought.

#4 · Cardiovascular Emergencies

What ECG pattern is classic for acute pericarditis?

Quick Reference

Diffuse concave-up ST elevation across most leads along with PR segment depression, without the reciprocal changes seen in STEMI. The pain is typically pleuritic and improves when the patient leans forward. Bedside echo can screen for an associated effusion.

#5 · Cardiovascular Emergencies

A patient develops sudden chest pain and subcutaneous emphysema after forceful vomiting. What diagnosis must be considered?

Quick Reference

Boerhaave syndrome, a spontaneous esophageal rupture from a sudden rise in intraluminal pressure. Look for Hamman's crunch on auscultation and mediastinal air on imaging. This is a surgical emergency because untreated mediastinitis carries a very high mortality rate.

#6 · Cardiovascular Emergencies

What clinical tool helps estimate pretest probability of pulmonary embolism at the bedside?

Quick Reference

The Wells score, which weighs factors like tachycardia, immobilization or recent surgery, prior VTE, hemoptysis, and whether PE is the most likely diagnosis. A low score combined with a negative PERC rule can safely avoid further testing in low-risk patients. Higher scores push toward D-dimer or direct CT pulmonary angiography.

#7 · Cardiovascular Emergencies

Why is reproducible chest wall tenderness on palpation not enough to rule out a cardiac cause?

Quick Reference

Musculoskeletal-sounding pain that is reproducible on palpation can coexist with true ACS, and relying on this finding alone is a common trap. Risk factors, ECG, and troponin should still guide the workup in at-risk patients. Costochondritis is a diagnosis of exclusion, not a first assumption.

#8 · Cardiovascular Emergencies

What do the letters in the HEART score stand for, and what is it used for?

Quick Reference

History, ECG, Age, Risk factors, and Troponin. It stratifies emergency department chest pain patients into low, moderate, and high risk categories to guide safe early discharge versus admission for further cardiac workup. A low HEART score supports outpatient follow-up rather than admission.

#9 · Cardiovascular Emergencies

What is the classic triad of findings in tension pneumothorax, and what is the immediate treatment?

Quick Reference

Jugular venous distension, absent breath sounds on the affected side, and tracheal deviation away from the affected side, usually with hypotension. This is a clinical diagnosis that should not wait for imaging. Immediate needle decompression or finger thoracostomy is required before a chest tube is placed.

#10 · Cardiovascular Emergencies

Why should beta blockers be avoided in cocaine-induced chest pain?

Quick Reference

Beta blockade leaves alpha-adrenergic stimulation unopposed, which can worsen coronary vasospasm and hypertension. Benzodiazepines and nitrates are the preferred first-line agents instead. This is a frequently tested exception to standard ACS management.

#11 · Cardiovascular Emergencies

Does a normal or non-diagnostic ECG rule out acute coronary syndrome?

Quick Reference

No. Up to a significant proportion of true ACS presentations have a normal or nonspecific initial ECG, especially early in the course. Serial ECGs and serial troponins over time are needed to safely exclude ACS in a patient with a concerning history. A single normal ECG should never be used alone to discharge a high-risk patient.

#12 · Cardiovascular Emergencies

Why can nitrate responsiveness be misleading when evaluating chest pain?

Quick Reference

Esophageal spasm can also relieve with nitrates, so pain improvement after nitroglycerin does not confirm a cardiac cause. This is a classic exam trap where the test taker assumes a positive nitrate response equals ACS. Diagnosis still requires ECG and troponin correlation.

#13 · Cardiovascular Emergencies

What less common but important causes of chest pain should be considered in young, otherwise healthy patients?

Quick Reference

Myocarditis, spontaneous pneumothorax in tall thin males, and cocaine or other stimulant use should all be on the differential even without typical cardiac risk factors. Anxiety and musculoskeletal pain are common but should remain diagnoses of exclusion. A careful history around substance use and recent viral illness is important.

#14 · Cardiovascular Emergencies

Why are right-sided ECG leads obtained in a patient with an inferior STEMI, and what treatment must be avoided if they are positive?

Quick Reference

Right-sided leads (especially V4R) detect right ventricular infarction, which complicates a significant proportion of inferior MIs. An RV infarct is preload-dependent, so nitrates and diuretics can cause severe hypotension and must be avoided. These patients are instead managed with fluid boluses to maintain preload.

#15 · Cardiovascular Emergencies

What are the ECG criteria that define a STEMI?

Quick Reference

New ST elevation at the J point in two anatomically contiguous leads, meeting specific millimeter thresholds that vary by lead location and age or sex, or a new left bundle branch block with concordant ST changes. These findings should trigger urgent reperfusion pathways. Reciprocal ST depression in other leads supports the diagnosis.

#16 · Cardiovascular Emergencies

What are the target reperfusion times for STEMI treatment?

Quick Reference

Door-to-balloon time of 90 minutes or less for primary PCI, or door-to-needle time of 30 minutes or less for fibrinolysis when timely PCI is not available. If PCI cannot be achieved within about 120 minutes of first medical contact, fibrinolysis should be given instead. Time to reperfusion directly affects myocardial salvage.

What are the Sgarbossa criteria used for?

Quick Reference

They help identify acute MI in the presence of a left bundle branch block, where standard ST elevation criteria do not apply. Criteria include concordant ST elevation, concordant ST depression in V1 to V3, and excessively discordant ST elevation. A positive score supports emergent reperfusion even though the baseline ECG is already abnormal.

How are NSTEMI and unstable angina distinguished from each other?

Quick Reference

Both present with similar ischemic symptoms and possible ECG changes such as ST depression or T wave inversion, but NSTEMI has a positive troponin indicating myocardial necrosis while unstable angina does not. Management is similar initially, but troponin trend guides risk stratification and disposition. Both fall under the non-ST elevation ACS spectrum.

#19 · Cardiovascular Emergencies

What is the initial medical management bundle for suspected ACS in the emergency department?

Quick Reference

Chewed acetylsalicylic acid, a P2Y₁₂ inhibitor, anticoagulation such as heparin, and nitrates unless contraindicated by hypotension, right ventricular infarction, or recent phosphodiesterase-5 inhibitor use. Supplemental oxygen is given only if the patient is hypoxic, not routinely. Pain control and continuous cardiac monitoring round out initial care.

#20 · Cardiovascular Emergencies

What are key contraindications to fibrinolytic therapy in STEMI?

Quick Reference

Any history of hemorrhagic stroke, active internal bleeding, recent major surgery or trauma, and severely uncontrolled hypertension are major contraindications. These patients need transfer for primary PCI instead. Careful screening avoids catastrophic intracranial hemorrhage from thrombolytics.

#21 · Cardiovascular Emergencies

A post-MI patient develops sudden cardiogenic shock with a new holosystolic murmur. What mechanical complications should be considered?

Quick Reference

Papillary muscle rupture causing acute mitral regurgitation, or a ventricular septal defect from septal rupture. Both present with a new murmur and rapid hemodynamic deterioration and require urgent echocardiography and surgical consultation. Free wall rupture is another mechanical complication but usually presents with tamponade rather than a murmur.

#22 · Cardiovascular Emergencies

What is the typical timing of major mechanical complications after myocardial infarction?

Quick Reference

Free wall rupture and papillary muscle rupture classically occur within the first few days to about a week after infarction, when necrotic myocardium is weakest. Dressler syndrome, an autoimmune pericarditis, occurs weeks to months later. Recognizing this timeline helps explain sudden deterioration in a recovering MI patient.

#23 · Cardiovascular Emergencies

What is the classic presentation of right ventricular infarction, and how is it managed differently?

Quick Reference

Hypotension with clear lung fields and elevated jugular venous pressure, usually accompanying an inferior STEMI. Unlike left-sided failure, treatment relies on fluid boluses to maintain right-sided preload rather than diuretics or nitrates. Giving nitrates here can precipitate severe hypotension.

#24 · Cardiovascular Emergencies

Why should intravenous beta blockers be used cautiously in the acute phase of MI?

Quick Reference

Early aggressive IV beta blockade can worsen outcomes in patients with signs of heart failure, low cardiac output, or cardiogenic shock risk. Oral beta blockers are typically started once the patient is hemodynamically stable. This is a shift from older teaching that recommended routine early IV dosing.

#25 · Cardiovascular Emergencies

What secondary prevention medications are started after an MI before discharge?

Quick Reference

Dual antiplatelet therapy, a high-intensity statin, an ACE inhibitor or ARB, and a beta blocker in most patients, along with risk factor modification counseling. These reduce recurrent events and improve long-term survival. Timing and choice may vary based on comorbidities and revascularization strategy.

#26 · Cardiovascular Emergencies

How does pretest probability influence the choice between stress testing and cardiac catheterization for suspected stable ischemic heart disease?

Quick Reference

Low pretest probability patients may be reassured with a normal stress test, while intermediate probability patients often need further functional or anatomic testing. High pretest probability patients with concerning features are more often referred directly for catheterization. Risk factor burden and symptom character both feed into this decision.

#27 · Cardiovascular Emergencies

How is a posterior wall MI identified on a standard 12-lead ECG?

Quick Reference

It appears as ST depression and tall, broad R waves in leads V1 to V3, essentially a mirror image of anterior ST elevation. Posterior leads V7 to V9 can confirm true ST elevation over the posterior wall. This is a STEMI equivalent that is easy to miss if only the standard 12 leads are reviewed.

#28 · Cardiovascular Emergencies

Why might diabetic patients with ischemic heart disease present without the classic anginal symptoms?

Quick Reference

Autonomic neuropathy from long-standing diabetes can blunt cardiac pain perception, leading to silent ischemia or atypical symptoms such as dyspnea, fatigue, or diaphoresis alone. Clinicians must maintain a low threshold to obtain an ECG and troponin in diabetic patients with vague symptoms. This atypical presentation contributes to delayed diagnosis in this population.

#29 · Cardiovascular Emergencies

What is the first-line treatment for an unstable patient with a tachyarrhythmia?

Quick Reference

Immediate synchronized cardioversion, regardless of the specific rhythm, when the patient shows signs of instability such as hypotension, altered mental status, chest pain, or heart failure. Sedation should be given if time permits but should not delay cardioversion in a truly unstable patient. Pulseless VT or VF instead requires unsynchronized defibrillation.

#30 · Cardiovascular Emergencies

What is the stepwise approach to a stable, regular, narrow-complex tachycardia?

Quick Reference

Start with vagal maneuvers such as the modified Valsalva technique, and if unsuccessful move to adenosine given as a rapid IV push followed by a saline flush. If adenosine fails or the rhythm recurs, rate-controlling agents like a calcium channel blocker or beta blocker can be considered. Continuous ECG monitoring during administration is essential to capture the rhythm at the moment of conversion.

#31 · Cardiovascular Emergencies

Why might adenosine be used cautiously in patients with reactive airway disease?

Quick Reference

Adenosine can theoretically provoke bronchospasm in patients with asthma or COPD, so calcium channel blockers like verapamil or diltiazem are often preferred alternatives in this population. In practice adenosine is still frequently used given its short half-life, but caution and monitoring are advised. This nuance is worth knowing for exam purposes even though real-world practice varies.

#32 · Cardiovascular Emergencies

How should a wide-complex tachycardia of uncertain origin be treated in an older patient with known structural heart disease?

Quick Reference

It should be treated as ventricular tachycardia until proven otherwise, since VT is far more common than SVT with aberrancy in this population. Adenosine or verapamil should be avoided if VT cannot be excluded, since calcium channel blockers can cause hemodynamic collapse in true VT. Amiodarone or procainamide are appropriate options if the patient is stable.

#33 · Cardiovascular Emergencies

How does management of atrial fibrillation with rapid ventricular response differ between stable and unstable patients?

Quick Reference

Unstable patients need immediate synchronized cardioversion, while stable patients are typically managed with rate control using beta blockers or calcium channel blockers. Anticoagulation status and duration of AF onset affect whether cardioversion is safe from a stroke risk standpoint. The underlying trigger, such as sepsis or thyrotoxicosis, should also be sought and treated.

#34 · Cardiovascular Emergencies

Why are AV nodal blocking agents like calcium channel blockers dangerous in pre-excited atrial fibrillation from WPW?

Quick Reference

Blocking the AV node preferentially can promote conduction down the accessory pathway, leading to extremely rapid ventricular rates and degeneration into ventricular fibrillation. Procainamide is the preferred antiarrhythmic in this setting instead. The irregular, wide-complex, and often very fast rhythm on ECG should raise suspicion for this entity.

#35 · Cardiovascular Emergencies

What is torsades de pointes and how is it treated?

Quick Reference

It is a polymorphic ventricular tachycardia associated with a prolonged QT interval, often triggered by QT-prolonging drugs or electrolyte abnormalities. First-line treatment is IV magnesium sulfate even if the serum magnesium level is normal. Any offending medications should be stopped and electrolytes corrected.

#36 · Cardiovascular Emergencies

What is the initial drug treatment for unstable symptomatic bradycardia?

Quick Reference

Atropine is first line, given as repeated IV boluses. If atropine fails, transcutaneous pacing, or an infusion of dopamine or epinephrine, should follow while preparing for transvenous pacing. The underlying cause, such as medication toxicity or ischemia, should also be addressed.

#37 · Cardiovascular Emergencies

Why is atropine often ineffective in high-degree AV block?

Quick Reference

Atropine works by blocking vagal tone at the AV node, but Mobitz type II and third-degree (complete) AV block are usually caused by disease below the AV node that atropine cannot overcome. These patients should move straight to transcutaneous pacing rather than waiting on repeated atropine doses. Delaying pacing while giving ineffective atropine is a common management error.

#38 · Cardiovascular Emergencies

What ECG changes progress as serum potassium rises in hyperkalemia, and what is given first to stabilize the heart?

Quick Reference

Peaked T waves appear first, followed by PR prolongation and P wave flattening, then a widened QRS, and finally a sine wave pattern preceding cardiac arrest. IV calcium gluconate or calcium chloride is given first to stabilize the cardiac membrane, even though it does not lower potassium. Insulin with glucose, and other potassium-shifting or eliminating therapies, follow to actually lower the level.

#39 · Cardiovascular Emergencies

What arrhythmia is classically associated with digoxin toxicity, and how is severe toxicity treated?

Quick Reference

Digoxin can cause nearly any arrhythmia, but bidirectional ventricular tachycardia is considered a classic and fairly specific finding. Severe toxicity with hemodynamic instability or life-threatening arrhythmia is treated with digoxin-specific antibody fragments (digoxin immune Fab). Hyperkalemia in the setting of digoxin toxicity should be treated cautiously since calcium administration was historically felt to worsen toxicity, though this is now debated.

#40 · Cardiovascular Emergencies

What classes of medications commonly prolong the QT interval and increase torsades risk?

Quick Reference

Certain antipsychotics, macrolide antibiotics, and antiemetics such as ondansetron are frequently implicated in QT prolongation. Combining two or more QT-prolonging drugs, or giving them to a patient with baseline long QT syndrome or electrolyte abnormalities, raises the risk further. A baseline ECG is reasonable before starting these drugs in at-risk patients.

#41 · Cardiovascular Emergencies

What is the immediate treatment for pulseless ventricular tachycardia or ventricular fibrillation?

Quick Reference

Immediate unsynchronized defibrillation combined with high-quality CPR, following the standard ACLS cardiac arrest algorithm. Epinephrine and antiarrhythmics such as amiodarone are given per protocol between shocks. Reversible causes should be sought throughout the resuscitation.

#42 · Cardiovascular Emergencies

What is sick sinus syndrome and how does it typically present?

Quick Reference

It is dysfunction of the sinoatrial node leading to alternating episodes of bradycardia and tachycardia, often called tachy-brady syndrome. Patients may present with syncope, palpitations, or fatigue related to these swings. Definitive treatment is usually a permanent pacemaker, sometimes combined with rate-controlling medication for the tachycardic episodes.

#43 · Cardiovascular Emergencies

What is the CHA2DS2-VASc score used for?

Quick Reference

It estimates stroke risk in patients with atrial fibrillation to guide the decision about starting long-term anticoagulation. Points are assigned for congestive heart failure, hypertension, age, diabetes, prior stroke or TIA, vascular disease, and sex category. A higher score supports anticoagulation to reduce thromboembolic stroke risk.

#44 · Cardiovascular Emergencies

What is the classic clinical presentation of cardiogenic pulmonary edema?

Quick Reference

Orthopnea, paroxysmal nocturnal dyspnea, bilateral basilar crackles, and often a third heart sound on auscultation. Patients may also have jugular venous distension and peripheral edema if right-sided failure coexists. Rapid onset with severe hypertension suggests a flash pulmonary edema picture rather than gradual fluid accumulation.

What is SCAPE and how is it treated?

Quick Reference

Sympathetic Crashing Acute Pulmonary Edema is sudden severe pulmonary edema driven by a hypertensive surge with fluid redistribution rather than pure volume overload. Treatment centers on high-dose nitroglycerin and non-invasive ventilation, with minimal or no diuretic since these patients are often not volume overloaded. Rapid afterload reduction is the priority rather than aggressive diuresis.

Why is furosemide not the primary first-line agent in flash pulmonary edema with a hypertensive patient?

Quick Reference

In flash pulmonary edema, fluid has often redistributed centrally from peripheral vasoconstriction rather than accumulated from true volume excess, so afterload reduction with nitrates matters more initially. Diuretics still have a role but are secondary to nitrates and non-invasive ventilation in this specific presentation. Relying on furosemide alone as the first move is a common management error.

#47 · Cardiovascular Emergencies

What contraindications limit the use of non-invasive ventilation in pulmonary edema?

Quick Reference

Decreased level of consciousness with inability to protect the airway, active vomiting with aspiration risk, and untreated pneumothorax are key contraindications. In these situations, intubation should be pursued instead of persisting with a mask interface. Facial trauma or inability to fit the mask properly are additional practical barriers.

#48 · Cardiovascular Emergencies

What are common precipitants of acute decompensated heart failure that should be sought in every patient?

Quick Reference

Medication or dietary sodium non-adherence, myocardial ischemia, new arrhythmia such as atrial fibrillation, and infection are frequent triggers. Identifying and treating the precipitant is as important as treating the pulmonary edema itself. Anemia and uncontrolled hypertension are additional contributors to consider.

#49 · Cardiovascular Emergencies

What limits the diagnostic usefulness of BNP or NT-proBNP in suspected heart failure?

Quick Reference

Levels rise with renal impairment even without true volume overload, and can be falsely lowered in obese patients due to increased clearance by adipose tissue. A single value should be interpreted alongside the clinical picture and imaging rather than used in isolation. Trend over time can still be useful for monitoring response to treatment.

#50 · Cardiovascular Emergencies

What chest x-ray findings support a diagnosis of cardiogenic pulmonary edema?

Quick Reference

Cephalization of pulmonary vessels, Kerley B lines from interstitial edema, a perihilar batwing pattern of alveolar edema, and often bilateral pleural effusions. Cardiomegaly may also be present if there is chronic underlying heart failure. These findings help distinguish cardiogenic edema from other causes of bilateral infiltrates.

#51 · Cardiovascular Emergencies

Why has morphine fallen out of favor as routine treatment for acute pulmonary edema?

Quick Reference

Observational data linked morphine use in acute heart failure to worse outcomes, including increased need for mechanical ventilation and higher mortality. It is no longer recommended as a routine adjunct and should be reserved for specific symptom control needs rather than standard therapy. Nitrates and non-invasive ventilation remain the mainstays instead.

#52 · Cardiovascular Emergencies

How do the physical exam findings of left-sided and right-sided heart failure differ?

Quick Reference

Left-sided failure causes pulmonary symptoms such as crackles, orthopnea, and paroxysmal nocturnal dyspnea from backup into the lungs. Right-sided failure causes systemic congestion with elevated jugular venous pressure, hepatomegaly, and peripheral edema. Many patients with chronic heart failure have overlapping biventricular findings.

#53 · Cardiovascular Emergencies

What signs indicate that a pulmonary edema patient needs intubation rather than continued non-invasive ventilation?

Quick Reference

Failure to improve or worsening respiratory status despite adequate non-invasive ventilation, respiratory muscle fatigue, worsening mental status, or hemodynamic instability all point toward the need for intubation. Non-invasive ventilation should not be persisted with indefinitely if the patient is clearly deteriorating. Early recognition avoids a crash intubation in an unstable patient.

#54 · Cardiovascular Emergencies

What valvular lesions classically cause sudden flash pulmonary edema?

Quick Reference

Acute severe mitral regurgitation, for example from papillary muscle rupture, and severe aortic stenosis with a sudden hemodynamic stressor are classic valvular causes. These lesions can produce dramatic pulmonary edema even without chronic heart failure history. Bedside echocardiography can rapidly identify the underlying valve pathology.

#55 · Cardiovascular Emergencies

What workup should follow stabilization of a first episode of pulmonary edema?

Quick Reference

An ECG and troponin to look for underlying ischemia, electrolytes and renal function, and an echocardiogram to assess ejection fraction and valvular function. Identifying whether the cause is ischemic, valvular, hypertensive, or arrhythmic guides both acute and long-term management. This workup should not be skipped just because the patient responds well to initial therapy.

#56 · Cardiovascular Emergencies

How can point-of-care lung ultrasound assist in diagnosing pulmonary edema at the bedside?

Quick Reference

Multiple bilateral B-lines on lung ultrasound support a diagnosis of pulmonary edema and can help distinguish it from other causes of dyspnea such as pneumothorax, which instead shows absent lung sliding. This bedside tool can speed diagnosis before formal imaging is available. It is particularly useful in patients too unstable for transport to radiology.

#57 · Cardiovascular Emergencies

Why is controlled, gradual diuresis sometimes preferred over aggressive diuresis in chronic decompensated heart failure?

Quick Reference

Overly aggressive diuresis can cause acute kidney injury and electrolyte derangements, and does not address the afterload problem that is often the main driver in hypertensive flash pulmonary edema. A combination of vasodilator therapy and judicious diuresis, guided by the patient's volume status, tends to produce better outcomes. Daily weight and renal function monitoring help titrate therapy safely.

#58 · Respiratory Emergencies

What clinical findings define a severe asthma or COPD exacerbation?

Quick Reference

Marked accessory muscle use, inability to speak in full sentences, a peak expiratory flow rate less than 50 percent of predicted, and significant tachypnea or tachycardia all indicate a severe exacerbation. Oxygen saturation may still look reassuring early on despite significant distress. These findings should prompt aggressive early treatment rather than a wait-and-see approach.

#59 · Respiratory Emergencies

Why is a silent chest on auscultation an ominous finding in asthma, not a reassuring one?

Quick Reference

A silent chest usually means airflow is so severely reduced that wheezing can no longer be generated, signaling impending respiratory failure rather than improvement. This finding should prompt urgent escalation of therapy and preparation for possible intubation. Clinicians must not mistake the absence of wheeze for clinical improvement.

#60 · Respiratory Emergencies

What is the standard first-line treatment bundle for an asthma or COPD exacerbation?

Quick Reference

Short-acting beta agonists, ipratropium bromide, and early systemic corticosteroids form the core of initial treatment. Corticosteroids should be given early since their onset of action takes hours, so delaying them prolongs the exacerbation. Oxygen is titrated to target saturation rather than given indiscriminately, especially in COPD.

#61 · Respiratory Emergencies

What adjunct therapy is used for severe asthma exacerbations that do not respond adequately to standard bronchodilators?

Quick Reference

Intravenous magnesium sulfate can provide additional bronchodilation in severe asthma unresponsive to initial beta agonist and anticholinergic therapy. It is generally reserved for more severe presentations rather than used routinely in mild exacerbations. Continuous nebulized bronchodilators may also be used in this setting.

#62 · Respiratory Emergencies

Why is non-invasive ventilation used with more caution in severe asthma compared to COPD exacerbations?

Quick Reference

Severe asthma carries a risk of breath stacking and auto-PEEP from prolonged expiratory time needs, and positive pressure ventilation can worsen dynamic hyperinflation if not carefully managed. In COPD, non-invasive ventilation is well established as first-line for exacerbations with respiratory acidosis. Close monitoring is essential if non-invasive ventilation is attempted in a severe asthmatic.

#63 · Respiratory Emergencies

What ventilator strategy is used to avoid breath stacking in a severely asthmatic patient who requires intubation?

Quick Reference

Permissive hypercapnia with a lower respiratory rate and prolonged expiratory time allows full exhalation and avoids progressive air trapping. Tidal volumes are also kept modest to limit peak airway pressures. Accepting a higher carbon dioxide level is considered safer than risking barotrauma or hemodynamic collapse from severe auto-PEEP.

#64 · Respiratory Emergencies

What is the most common trigger for an acute COPD exacerbation?

Quick Reference

Respiratory tract infection, whether viral or bacterial, is the most common precipitant. Other important triggers to consider include pulmonary embolism, pneumothorax, medication non-adherence, and environmental irritant exposure. Identifying the trigger helps direct additional targeted therapy beyond standard bronchodilators and steroids.

#65 · Respiratory Emergencies

What is the target oxygen saturation range in a COPD patient, and why does it matter?

Quick Reference

A controlled target of roughly 88 to 92 percent is generally used rather than driving saturation to near 100 percent. Excess oxygen can worsen hypercapnia through altered ventilation-perfusion matching and reduced hypoxic respiratory drive in chronic CO₂ retainers. Careful oxygen titration avoids precipitating or worsening hypercapnic respiratory failure.

#66 · Respiratory Emergencies

What is the first-line respiratory support for a COPD exacerbation with respiratory acidosis on blood gas?

Quick Reference

Non-invasive positive pressure ventilation, typically BiPAP, is first-line and has strong evidence for reducing the need for intubation in this setting. It should be started early rather than reserved as a last resort before intubation. Contraindications such as decreased level of consciousness or inability to protect the airway must be excluded first.

#67 · Respiratory Emergencies

When are antibiotics indicated in an acute COPD exacerbation?

Quick Reference

Antibiotics are indicated when there is increased sputum purulence along with increased sputum volume or worsening dyspnea, or in patients requiring mechanical ventilation. Not every COPD exacerbation needs antibiotics, since many are viral in origin. Choice of antibiotic considers local resistance patterns and prior exacerbation history.

#68 · Respiratory Emergencies

What peak expiratory flow rate threshold generally defines a severe asthma exacerbation?

Quick Reference

A peak flow less than 50 percent of the patient's predicted or personal best value is generally considered severe. Values below this threshold usually warrant more aggressive treatment and closer observation before considering discharge. Serial peak flow measurements also help track response to treatment over the emergency department visit.

#69 · Respiratory Emergencies

Why can giving high concentrations of supplemental oxygen worsen hypercapnia in a chronic CO₂ retainer?

Quick Reference

High oxygen can blunt hypoxic respiratory drive in patients dependent on it, worsen ventilation-perfusion mismatch by reversing hypoxic pulmonary vasoconstriction, and increase hemoglobin's affinity for CO₂ release through the Haldane effect. The combined result is rising CO₂ levels and worsening acidosis. This is why oxygen is titrated carefully rather than maximized in COPD.

#70 · Respiratory Emergencies

Why should sudden clinical deterioration in a patient with severe asthma or COPD prompt evaluation for pneumothorax?

Quick Reference

Both conditions predispose to spontaneous pneumothorax from bullous disease or barotrauma during aggressive ventilation. A sudden drop in oxygen saturation, new hypotension, or asymmetric breath sounds should prompt urgent evaluation. This diagnosis is easy to miss if attributed solely to worsening bronchospasm.

#71 · Respiratory Emergencies

What criteria support safe discharge after treatment for an asthma exacerbation?

Quick Reference

Sustained clinical improvement after the last bronchodilator treatment, a peak flow greater than about 70 percent of predicted, and a clear follow-up and medication plan including confirmed inhaler technique. Patients discharged too early without addressing technique or adherence are at higher risk of early relapse. A short course of oral steroids is usually continued after discharge.

#72 · Respiratory Emergencies

What broad categories should structure the differential diagnosis of undifferentiated dyspnea?

Quick Reference

Cardiac causes, pulmonary causes, and a third category including anemia, metabolic acidosis, anxiety, and neuromuscular weakness. Organizing the differential this way prevents anchoring on the lungs or heart alone. A thorough history and targeted exam help narrow which category is most likely.

#73 · Respiratory Emergencies

What is the PERC rule used for in dyspnea and chest pain evaluation?

Quick Reference

The Pulmonary Embolism Rule-out Criteria is a set of eight clinical variables that, if all negative in a low pretest probability patient, allow PE to be excluded without further testing. It is intended only for patients already judged low risk by gestalt or a formal score like Wells. Using PERC in a higher risk patient inappropriately could miss a true PE.

#74 · Respiratory Emergencies

In whom should D-dimer testing be used to help exclude pulmonary embolism?

Quick Reference

D-dimer has good sensitivity but poor specificity, so it is most useful in patients with a low or intermediate pretest probability, where a negative result can safely exclude PE. In high pretest probability patients, D-dimer should not be relied upon and imaging should proceed directly. Age-adjusted D-dimer thresholds can improve specificity in older patients.

#75 · Respiratory Emergencies

What exam findings suggest pneumothorax as a cause of acute dyspnea?

Quick Reference

Hyperresonance to percussion, decreased or absent breath sounds on the affected side, and in tension physiology, tracheal deviation away from the affected side with hypotension. Sudden onset pleuritic pain often accompanies the dyspnea. Lung point on ultrasound or absent lung sliding can confirm the diagnosis at the bedside.

#76 · Respiratory Emergencies

What is the first-line treatment for dyspnea caused by anaphylaxis?

Quick Reference

Intramuscular epinephrine into the anterolateral thigh is first-line and should not be delayed while other treatments are prepared. Antihistamines and corticosteroids are adjuncts only and do not replace epinephrine as initial therapy. Airway swelling can progress rapidly, so early airway assessment is also critical.

#77 · Respiratory Emergencies

Why should hemoglobin be checked in a patient with unexplained dyspnea and no obvious cardiopulmonary cause?

Quick Reference

Significant anemia reduces oxygen-carrying capacity and can present primarily as dyspnea on exertion or even at rest, especially if it developed gradually. A normal cardiopulmonary workup should prompt consideration of hematologic causes. Transfusion or treatment of the underlying cause of anemia resolves the dyspnea in these cases.

#78 · Respiratory Emergencies

Why does diabetic ketoacidosis cause a change in breathing pattern, and what is this pattern called?

Quick Reference

Kussmaul breathing is deep, labored breathing that represents respiratory compensation for the underlying metabolic acidosis, not primary lung disease. Recognizing this pattern should prompt checking a glucose and blood gas rather than focusing solely on the chest. Treating the underlying DKA resolves the abnormal breathing pattern.

#79 · Respiratory Emergencies

What historical and exam clues raise suspicion for foreign body aspiration causing dyspnea?

Quick Reference

A sudden choking episode, especially during eating or play in a young child, followed by cough, stridor, or unilateral wheeze should raise suspicion. Chest x-ray may show unilateral hyperinflation or, less commonly, a radiopaque object. Bronchoscopy is both diagnostic and therapeutic when foreign body aspiration is strongly suspected.

#80 · Respiratory Emergencies

Why is panic disorder or hyperventilation syndrome considered a diagnosis of exclusion in dyspnea?

Quick Reference

Anxiety-related hyperventilation can mimic serious organic causes of dyspnea, including tingling and chest tightness, but a full workup should exclude cardiac, pulmonary, and metabolic causes first. Labeling dyspnea as purely anxiety-related without adequate workup risks missing a true emergency. Only after other causes are reasonably excluded should this diagnosis be made.

#81 · Respiratory Emergencies

How does point-of-care lung ultrasound help differentiate causes of acute dyspnea?

Quick Reference

Multiple bilateral B-lines suggest pulmonary edema, absent lung sliding suggests pneumothorax, and consolidation patterns can suggest pneumonia. This bedside tool can rapidly narrow the differential when combined with the clinical picture. It is especially useful in unstable patients who cannot be moved for formal imaging.

#82 · Respiratory Emergencies

What neuromuscular conditions should be considered in a patient with dyspnea and normal-appearing lungs?

Quick Reference

Myasthenic crisis and Guillain-Barre syndrome can cause respiratory muscle weakness leading to dyspnea and eventual respiratory failure without primary lung pathology. Bedside measurement of negative inspiratory force and vital capacity helps track deterioration and guide the need for intubation. These conditions can progress rapidly and are easy to underestimate early on.

#83 · Respiratory Emergencies

Why is stridor considered a red flag symptom in a dyspneic patient?

Quick Reference

Stridor indicates upper airway obstruction, as seen in epiglottitis, severe angioedema, or foreign body, and can progress rapidly to complete airway obstruction. Early recognition and preparation for a difficult airway, sometimes involving anesthesia or otolaryngology, is critical. Attempting routine airway maneuvers without this preparation can worsen the obstruction.

#84 · Respiratory Emergencies

What exam finding suggests a pleural effusion as the cause of dyspnea, and what confirms it?

Quick Reference

Dullness to percussion with decreased breath sounds over the affected area suggests effusion, and bedside ultrasound or chest x-ray confirms the diagnosis and estimates volume. Large effusions may require therapeutic thoracentesis for symptom relief. The underlying cause, whether infectious, malignant, or heart failure related, still needs to be investigated.

#85 · Respiratory Emergencies

Why might an elderly patient with acute coronary syndrome present with dyspnea rather than chest pain?

Quick Reference

Atypical presentation is common in older adults, who may report only shortness of breath, fatigue, or weakness rather than classic angina. This atypical picture contributes to delayed recognition of ACS in this age group. A low threshold for ECG and troponin testing is warranted in elderly patients with unexplained dyspnea.

#86 · Neurological Emergencies

What is the standard time window for administering IV thrombolysis in acute ischemic stroke?

Quick Reference

Within 4.5 hours of the patient's last known well time, based on strict inclusion and exclusion criteria. Mechanical thrombectomy can be offered up to 24 hours in carefully selected patients with a large vessel occlusion and favorable imaging. Time from last known well, not time of discovery, is what defines the window.

#87 · Neurological Emergencies

What are key absolute contraindications to IV thrombolysis in stroke?

Quick Reference

Active internal bleeding, recent hemorrhagic stroke, recent major surgery, severe uncontrolled hypertension above the treatment threshold, and current therapeutic anticoagulation with an elevated INR are major contraindications. Each patient must be screened against the full contraindication checklist before treatment. Missing a contraindication can lead to catastrophic hemorrhage.

#88 · Neurological Emergencies

Why is a non-contrast CT head obtained before giving thrombolysis in suspected ischemic stroke?

Quick Reference

It rapidly identifies or excludes hemorrhage, since thrombolysis in a hemorrhagic stroke would be catastrophic. Early ischemic changes may be subtle or absent on CT, but the main purpose at this stage is ruling out bleeding rather than confirming ischemia. Advanced imaging such as CT angiography may follow to assess for large vessel occlusion.

#89 · Neurological Emergencies

What is the NIHSS used for in stroke assessment?

Quick Reference

The National Institutes of Health Stroke Scale quantifies the severity of neurological deficits using a standardized set of exam components. It helps guide treatment decisions, communicate severity between providers, and track changes over time. Higher scores generally indicate more severe strokes and larger territory involvement.

#90 · Neurological Emergencies

Why is posterior circulation stroke often missed or misdiagnosed as peripheral vertigo?

Quick Reference

It can present with vertigo, diplopia, ataxia, and dysarthria that overlap significantly with benign peripheral causes of dizziness. Subtle findings such as skew deviation, an abnormal head impulse test, or direction-changing nystagmus should raise suspicion for a central cause. A careful HINTS exam can help distinguish central from peripheral vertigo in acute vestibular syndrome.

#91 · Neurological Emergencies

What should always be checked first when a patient presents with a possible stroke, and why?

Quick Reference

Bedside glucose should be checked immediately, since hypoglycemia is a common and completely reversible stroke mimic. Other mimics include Todd's paralysis after a seizure, complex migraine, and conversion disorder. Correcting hypoglycemia can rapidly resolve neurological deficits that otherwise look identical to stroke.

#92 · Neurological Emergencies

What is a TIA, and what score helps estimate short-term stroke risk after one?

Quick Reference

A transient ischemic attack is a temporary episode of neurological dysfunction from ischemia without evidence of infarction on imaging. The ABCD2 score, incorporating age, blood pressure, clinical features, duration, and diabetes, helps stratify short-term risk of a subsequent stroke. Higher scores support more urgent workup and specialist follow-up.

#93 · Neurological Emergencies

How does blood pressure management differ between a patient receiving thrombolysis and one who is not?

Quick Reference

Patients being treated with thrombolysis need blood pressure controlled below a strict threshold, typically under 185 systolic and 110 diastolic, before and after treatment to reduce hemorrhage risk. Patients not receiving thrombolysis are usually managed with permissive hypertension, avoiding aggressive blood pressure lowering that could worsen cerebral perfusion. Hemorrhagic stroke instead calls for more aggressive blood pressure reduction.

#94 · Neurological Emergencies

What clinical signs suggest a large vessel occlusion that may benefit from mechanical thrombectomy?

Quick Reference

Gaze deviation toward the side of the lesion, severe neurological deficit, and a high NIHSS score all raise suspicion for large vessel occlusion. CT angiography confirms the occlusion and helps plan endovascular therapy. These patients should be triaged rapidly to a center capable of thrombectomy if it is not available locally.

#95 · Neurological Emergencies

How is anticoagulation reversed urgently in a patient on warfarin who develops a hemorrhagic stroke?

Quick Reference

Prothrombin complex concentrate is given for rapid factor replacement, along with IV vitamin K to sustain reversal. Fresh frozen plasma is an alternative but works more slowly and requires larger volumes. Rapid reversal is critical since ongoing anticoagulation allows the hematoma to continue expanding.

#96 · Neurological Emergencies

Why does a cerebellar stroke require close monitoring even if the initial presentation seems mild?

Quick Reference

Cerebellar infarcts can develop rapid edema that compresses the brainstem or causes obstructive hydrocephalus, leading to sudden deterioration. Early neurosurgical consultation is important since decompressive surgery may be needed if swelling progresses. A seemingly stable patient can decompensate quickly, so serial neurological checks are essential.

#97 · Neurological Emergencies

What clinical clues suggest carotid artery dissection as the cause of stroke in a younger patient?

Quick Reference

Neck pain, a preceding history of trauma or neck manipulation, and an associated Horner syndrome (ptosis, miosis, anhidrosis) on the affected side are classic clues. This diagnosis should be considered in stroke patients without typical vascular risk factors. CT or MR angiography of the neck confirms the dissection.

#98 · Neurological Emergencies

How is the last known well time determined and used in a wake-up stroke?

Quick Reference

The last known well time is defined as the last time the patient was witnessed to be neurologically normal, which for a wake-up stroke is usually the time they went to sleep. Advanced imaging techniques, such as MRI with diffusion-perfusion mismatch or CT perfusion, can sometimes identify patients who still qualify for thrombolysis or thrombectomy despite an unknown exact onset time. This allows some wake-up stroke patients to still receive time-sensitive therapy.

#99 · Neurological Emergencies

Why is a dysphagia screen performed before giving any oral intake after an acute stroke?

Quick Reference

Stroke frequently impairs swallowing coordination, and unrecognized dysphagia significantly increases the risk of aspiration pneumonia. A bedside swallow screen should be completed before any food, fluid, or oral medication is given. Patients who fail the screen may need a modified diet or alternate route for medications until formal swallowing assessment is done.

#100 · Neurological Emergencies

When is aspirin started in a patient with acute ischemic stroke?

Quick Reference

In patients who do not receive thrombolysis, aspirin is typically started within 24 to 48 hours of symptom onset. In patients who do receive thrombolysis, aspirin is generally delayed until at least 24 hours after treatment to reduce hemorrhage risk. Early aspirin use has been shown to modestly reduce recurrent stroke risk and mortality.

#101 · Neurological Emergencies

What headache red flags should prompt urgent further workup rather than symptomatic treatment alone?

Quick Reference

Systemic symptoms like fever, focal neurological deficits, thunderclap onset, new headache after age fifty, a change in headache pattern, positional worsening, and papilledema all warrant urgent evaluation. These features distinguish a potentially dangerous secondary headache from a primary headache disorder. A headache with any of these features should not be treated as simple migraine without further workup.

#102 · Neurological Emergencies

How is subarachnoid hemorrhage worked up in a patient with thunderclap headache?

Quick Reference

Non-contrast CT head is the first test, and when performed within about six hours of symptom onset with modern scanners it has very high sensitivity. If CT is negative but performed later, or if suspicion remains high, lumbar puncture looking for xanthochromia and a red cell count that does not clear between tubes is the next step. CT angiography may follow to identify an aneurysm if hemorrhage is confirmed.

#103 · Neurological Emergencies

How does xanthochromia help distinguish true subarachnoid hemorrhage from a traumatic lumbar puncture?

Quick Reference

Xanthochromia is a yellowish discoloration of cerebrospinal fluid from breakdown of red blood cells, which takes several hours to develop and would not be present from a fresh traumatic tap. A true subarachnoid hemorrhage also tends to show a red cell count that does not decrease significantly between the first and last collection tubes. Both findings together support a genuine hemorrhage rather than a procedural artifact.

#104 · Neurological Emergencies

What features suggest giant cell arteritis as the cause of headache in an older patient, and why is urgent treatment important?

Quick Reference

New headache in a patient over fifty, jaw claudication, scalp tenderness, and visual symptoms suggest giant cell arteritis, supported by an elevated ESR or CRP. Empiric high-dose corticosteroids should be started urgently without waiting for biopsy confirmation, since delay risks permanent vision loss. A temporal artery biopsy is still obtained to confirm the diagnosis, ideally within about two weeks of starting steroids.

#105 · Neurological Emergencies

What is the classic presentation of idiopathic intracranial hypertension?

Quick Reference

It typically affects young, obese women with daily headache, papilledema on fundoscopic exam, and often visual disturbances such as transient visual obscurations. CT or MRI is usually normal, and diagnosis is confirmed by an elevated opening pressure on lumbar puncture. Untreated, it can lead to permanent vision loss from chronic papilledema.

#106 · Neurological Emergencies

What historical clues suggest cervical artery dissection as a cause of headache and neck pain?

Quick Reference

Recent neck trauma, chiropractic neck manipulation, or connective tissue disorders can predispose to dissection presenting with headache, neck pain, and sometimes focal neurological deficits from associated stroke. This diagnosis should be considered in younger patients with these risk factors and unexplained headache or neck pain. CT or MR angiography of the neck is used for confirmation.

#107 · Neurological Emergencies

When should a lumbar puncture be delayed in favor of CT imaging first, in suspected meningitis?

Quick Reference

CT should precede LP if there are focal neurological deficits, altered level of consciousness, papilledema, new seizure, or immunocompromise, since these raise concern for a mass lesion where LP could cause herniation. In patients without these features, LP should proceed without delay for imaging, and empiric antibiotics should not be withheld while awaiting either test. Blood cultures should be drawn before antibiotics if this can be done quickly.

#108 · Neurological Emergencies

What are effective first-line treatments for an acute migraine attack in the emergency department?

Quick Reference

Triptans, NSAIDs, and dopamine antagonist antiemetics such as prochlorperazine or metoclopramide are all effective first-line options and can be combined. IV fluids may help if the patient is dehydrated from associated vomiting. Opioids are generally avoided as first-line therapy due to limited efficacy for migraine and risk of medication overuse.

#109 · Neurological Emergencies

What is medication overuse headache and how is it recognized?

Quick Reference

It is a chronic daily or near-daily headache that develops from frequent use of acute headache medications, including simple analgesics, triptans, or opioids. It should be suspected in patients with chronic headache who are using abortive medications many days per month. Management involves gradually withdrawing the overused medication along with preventive therapy.

#110 · Neurological Emergencies

What is the classic presentation of cluster headache and its first-line acute treatment?

Quick Reference

Severe unilateral periorbital or temporal pain accompanied by ipsilateral autonomic features such as lacrimation, conjunctival injection, rhinorrhea, or ptosis, occurring in clusters over weeks. High-flow oxygen by non-rebreather mask and subcutaneous or intranasal sumatriptan are first-line acute treatments. Attacks are typically short but extremely severe and recurrent.

#111 · Neurological Emergencies

What is pituitary apoplexy and what clinical features suggest it?

Quick Reference

It is sudden hemorrhage or infarction of the pituitary gland, often into a pre-existing adenoma, presenting with sudden severe headache, visual field defects from optic chiasm compression, and sometimes acute hypopituitarism with hypotension. It is a neurosurgical and endocrine emergency requiring urgent imaging and stress-dose steroids. Rapid recognition prevents life-threatening adrenal crisis and permanent vision loss.

#112 · Neurological Emergencies

Why should carbon monoxide poisoning be considered when multiple people from the same household present with headache?

Quick Reference

Carbon monoxide poisoning classically causes headache, nausea, and confusion, and can affect several occupants of the same home or building simultaneously, especially in cold weather with heating equipment issues. Standard pulse oximetry does not reliably detect carboxyhemoglobin, so co-oximetry or a specific carboxyhemoglobin level is needed. A cluster presentation from one location is a strong clue that should not be missed.

#113 · Neurological Emergencies

Why must blood pressure and urine protein be checked in a pregnant or postpartum patient presenting with headache?

Quick Reference

Preeclampsia and eclampsia can present with headache as a prominent symptom, sometimes without other classic signs yet apparent, and can occur up to several weeks postpartum. Elevated blood pressure with proteinuria supports the diagnosis, and magnesium sulfate is used both to prevent and treat eclamptic seizures. Missing this diagnosis in a headache workup can have serious maternal consequences.

#114 · Neurological Emergencies

When should cerebral venous sinus thrombosis be considered in a patient with headache, and how is it diagnosed?

Quick Reference

It should be considered in patients with risk factors such as pregnancy, oral contraceptive use, or a hypercoagulable state, presenting with headache that may be gradual or thunderclap in onset, sometimes with seizures or focal deficits. CT or MR venography is used to visualize the thrombosed sinus, since standard non-contrast CT can miss this diagnosis. Anticoagulation is the mainstay of treatment even in the presence of some hemorrhagic conversion.

How is status epilepticus defined?

Quick Reference

A single seizure lasting longer than five minutes, or recurrent seizures without full return to baseline consciousness between them. This shorter operational definition reflects that ongoing seizure activity becomes progressively harder to terminate the longer it continues. It should be treated as a time-sensitive emergency rather than waiting the traditional thirty-minute definition used historically.

What is the first-line drug treatment for active status epilepticus?

Quick Reference

Benzodiazepines, such as IV lorazepam or, if IV access is not available, intramuscular midazolam. Prompt administration is critical since delayed treatment reduces the likelihood of seizure termination. Repeat dosing may be needed if the first dose does not stop the seizure within a few minutes.

#117 · Neurological Emergencies

What are the typical second-line agents used when benzodiazepines fail to terminate status epilepticus?

Quick Reference

Fosphenytoin or phenytoin, levetiracetam, and valproic acid are all accepted second-line options, chosen based on patient factors like comorbidities and allergies. These are given as an IV load after benzodiazepine failure to establish longer-term seizure control. If seizures continue despite an adequate second-line agent, the patient has entered refractory status epilepticus.

#118 · Neurological Emergencies

How is refractory status epilepticus managed?

Quick Reference

The patient requires intubation and continuous anesthetic infusion, typically with propofol or midazolam, along with continuous EEG monitoring to confirm seizure suppression. Refractory status epilepticus carries significant morbidity and needs ICU-level care. Identifying and treating the underlying cause remains important throughout this escalation.

#119 · Neurological Emergencies

What baseline workup should be obtained in a patient with a new-onset seizure?

Quick Reference

Bedside glucose, electrolytes including sodium and calcium, and head imaging to look for a structural cause. Lumbar puncture should be added if there is fever or signs of meningismus suggesting infection. The extent of workup is tailored to the clinical scenario, with a first unprovoked seizure in a healthy adult sometimes needing less extensive testing than one with red flag features.

#120 · Neurological Emergencies

How does treatment of an eclamptic seizure differ from standard status epilepticus management?

Quick Reference

Magnesium sulfate is the specific treatment of choice for eclampsia, both to stop the seizure and to prevent recurrence, rather than relying primarily on benzodiazepines or phenytoin. Blood pressure control and delivery planning are also central to management. Recognizing eclampsia as the cause changes the entire treatment approach compared to other seizure etiologies.

#121 · Neurological Emergencies

What is the treatment approach for alcohol withdrawal seizures, and what complication should be monitored for afterward?

Quick Reference

Benzodiazepines are the treatment of choice for alcohol withdrawal seizures, similar to other causes of status epilepticus. Patients should be monitored closely for progression to delirium tremens, which can develop over the following one to three days and carries significant morbidity if untreated. Thiamine should also be given given the risk of coexisting nutritional deficiency.

#122 · Neurological Emergencies

What is Todd's paralysis and why can it be mistaken for stroke?

Quick Reference

It is a transient focal neurological deficit, most often weakness, that follows a seizure and typically resolves over minutes to hours as the brain recovers from the post-ictal state. Because the weakness can be unilateral and dramatic, it is often mistaken for an acute stroke if the preceding seizure was unwitnessed. A history of seizure activity, or previous episodes with a known pattern, helps distinguish it from true ischemic stroke.

#123 · Neurological Emergencies

What features help distinguish psychogenic non-epileptic seizures from true epileptic seizures?

Quick Reference

Psychogenic events often show waxing and waning movements, asynchronous limb activity, pelvic thrusting, side-to-side head movement, preserved or fluctuating awareness during the event, and a rapid return to full alertness afterward without a typical post-ictal confusion period. True epileptic seizures more often show stereotyped tonic-clonic activity with a clear post-ictal state. Video EEG monitoring is the gold standard for definitive diagnosis when the distinction is unclear.

#124 · Neurological Emergencies

How are simple and complex febrile seizures distinguished, and why does this distinction matter?

Quick Reference

A simple febrile seizure is generalized, lasts less than fifteen minutes, and does not recur within twenty-four hours, and typically requires no extensive workup beyond finding the source of fever. A complex febrile seizure has focal features, lasts longer than fifteen minutes, or recurs within the same illness, and warrants a more thorough evaluation. This distinction guides how aggressively further investigation for a structural or infectious cause is pursued.

#125 · Neurological Emergencies

Why should sodium and glucose be checked immediately in a seizing patient, beyond the routine workup?

Quick Reference

Severe hyponatremia and hypoglycemia are both reversible metabolic causes of seizure that can be corrected rapidly at the bedside. Missing these treatable causes while pursuing more elaborate imaging or neurologic workup delays definitive treatment. Point-of-care testing allows immediate correction while the rest of the evaluation proceeds.

#126 · Neurological Emergencies

What specific antidote is used for seizures caused by isoniazid overdose, and why does standard treatment often fail?

Quick Reference

Pyridoxine, also known as vitamin B6, is the specific antidote for isoniazid-induced seizures, since isoniazid depletes B6 and impairs GABA synthesis. Standard benzodiazepines alone are often ineffective without also correcting this underlying deficiency. High-dose pyridoxine should be given promptly once isoniazid toxicity is suspected as the seizure cause.

#127 · Neurological Emergencies

What key considerations affect discharge planning after a first unprovoked seizure in an adult?

Quick Reference

Driving restrictions per local regulations, arranging outpatient neurology follow-up, and a discussion about whether to start an anti-epileptic medication based on recurrence risk factors and imaging or EEG findings. Not every first seizure requires starting long-term medication immediately. Patient education about seizure precautions and safety at home is also part of discharge counseling.

#128 · Neurological Emergencies

What is the appropriate positioning and airway management approach for a patient in the post-ictal period?

Quick Reference

The patient should be placed in the recovery position to protect the airway and reduce aspiration risk while they remain drowsy or confused. Suction should be available in case of vomiting or excess secretions. Vital signs and neurological status should be monitored closely until the patient returns to baseline.

#129 · Altered Mental Status and Behavioral Emergencies

What does the AEIOU-TIPS mnemonic help organize in the assessment of altered level of consciousness?

Quick Reference

It covers Alcohol, Epilepsy or Electrolytes, Insulin or metabolic causes, Overdose, Uremia, Trauma, Infection, Psychiatric causes, and Stroke or Space-occupying lesion. Using this framework helps ensure a broad, systematic differential rather than anchoring on one likely cause. It is especially useful when the history is limited because the patient cannot provide one.

#130 · Altered Mental Status and Behavioral Emergencies

What GCS threshold generally raises concern about the need for airway protection?

Quick Reference

A Glasgow Coma Scale of eight or less is the traditional threshold prompting strong consideration of intubation to protect the airway. This is a general guideline rather than an absolute rule, since the trend and clinical trajectory matter as much as the single number. Airway reflexes and the ability to manage secretions should also be assessed directly.

#131 · Altered Mental Status and Behavioral Emergencies

Why is bedside glucose checked immediately in every patient with decreased level of consciousness?

Quick Reference

Hypoglycemia is a rapidly reversible and common cause of altered mental status that can mimic stroke, intoxication, or other serious pathology. A single fingerstick glucose can prevent significant delay in treating a completely correctable problem. This step should never be skipped regardless of how obvious another cause seems.

#132 · Altered Mental Status and Behavioral Emergencies

When is naloxone appropriate as an empiric trial in altered level of consciousness?

Quick Reference

When there is respiratory depression with a slow respiratory rate along with pinpoint pupils, suggesting opioid toxicity as the likely cause. Naloxone can be titrated carefully to restore adequate respiration without necessarily fully reversing all sedation, to avoid precipitating acute withdrawal or agitation. It should be given alongside continued airway monitoring rather than as a substitute for it.

#133 · Altered Mental Status and Behavioral Emergencies

What is the classic triad of Wernicke encephalopathy, and why should thiamine be given before glucose in at-risk patients?

Quick Reference

Confusion, ataxia, and ophthalmoplegia make up the classic triad, though all three are not always present together. Giving glucose before thiamine in a thiamine-deficient patient can precipitate or worsen Wernicke encephalopathy by accelerating thiamine depletion during glucose metabolism. Thiamine should therefore be given first or simultaneously in any patient at risk, such as those with alcohol use disorder or malnutrition.

#134 · Altered Mental Status and Behavioral Emergencies

What findings support a diagnosis of hepatic encephalopathy, and how is it treated?

Quick Reference

Asterixis (a flapping tremor with wrist extension), confusion, and an elevated serum ammonia level in a patient with known or suspected liver disease support this diagnosis. Lactulose is the mainstay of treatment, working by reducing ammonia absorption in the gut. Identifying and treating a precipitant, such as infection or gastrointestinal bleeding, is also important since these commonly trigger episodes.

Why can hypoglycemia mimic an acute stroke, and what is the practical implication?

Quick Reference

Severe hypoglycemia can cause focal neurological deficits, including hemiparesis and aphasia, that are clinically indistinguishable from an acute ischemic stroke. Checking glucose before proceeding with an extensive stroke workup or thrombolysis prevents inappropriate treatment. Prompt glucose correction can rapidly reverse the deficits if hypoglycemia is truly the cause.

What are the key features that distinguish anticholinergic, cholinergic, sympathomimetic, and sedative-hypnotic toxidromes?

Quick Reference

Anticholinergic toxicity causes dry skin, dilated pupils, tachycardia, and agitation or delirium. Cholinergic toxicity causes excessive secretions, miosis, bradycardia, and muscle fasciculations. Sympathomimetic toxicity causes agitation, tachycardia, hypertension, and diaphoresis, while sedative-hypnotic toxicity causes CNS depression with normal or small pupils and preserved vital signs relative to the depth of sedation.

#137 · Altered Mental Status and Behavioral Emergencies

Why can standard pulse oximetry be misleading in carbon monoxide poisoning as a cause of decreased LOC?

Quick Reference

Standard pulse oximetry cannot distinguish carboxyhemoglobin from normal oxyhemoglobin, so it can display a falsely normal or near-normal saturation despite significant hypoxia at the tissue level. A specific carboxyhemoglobin level via co-oximetry is needed to make the diagnosis. This is a classic trap where a normal pulse ox reading gives false reassurance.

#138 · Altered Mental Status and Behavioral Emergencies

What clinical features suggest myxedema coma as a cause of decreased level of consciousness, and how is it treated?

Quick Reference

Severe hypothyroidism presenting with decreased mental status, hypothermia, bradycardia, and hypotension in an often elderly patient suggests myxedema coma. Treatment includes IV levothyroxine, stress-dose corticosteroids to cover possible coexisting adrenal insufficiency, and passive rewarming. This is a rare but life-threatening endocrine emergency that is easy to overlook if hypothyroidism is not on the differential.

#139 · Altered Mental Status and Behavioral Emergencies

Why should non-convulsive status epilepticus be considered in a patient with persistent unexplained decreased LOC?

Quick Reference

This condition can cause ongoing altered mental status without the obvious motor convulsions typically associated with seizures, making it easy to miss on routine exam. It should be suspected when a patient remains unresponsive without another clear explanation, especially after a prior convulsive seizure that appeared to stop. EEG is required to confirm the diagnosis since clinical exam alone cannot reliably detect it.

#140 · Altered Mental Status and Behavioral Emergencies

Which patients with decreased LOC after a fall should raise particular concern for subdural or epidural hematoma?

Quick Reference

Elderly patients and those on anticoagulant or antiplatelet medications are at increased risk, and symptoms may be delayed by hours to days, especially with subdural hematomas from bridging vein injury. A history of even a minor fall in these populations warrants a lower threshold for head imaging. Reversal of anticoagulation may be urgently needed if a hemorrhage is found.

Why might sepsis present as altered level of consciousness without an obvious fever, particularly in the elderly?

Quick Reference

Elderly patients often have a blunted febrile response to infection, so altered mentation, hypotension, or generalized weakness may be the primary presenting features of sepsis rather than fever. A high index of suspicion is needed since delayed recognition worsens outcomes. Broad screening for infection, including urine, blood, and lung sources, should be considered in these patients.

What is the general stepwise approach to the unresponsive patient in the emergency department?

Quick Reference

Start with airway, breathing, and circulation assessment and stabilization, then check bedside glucose, and consider empiric administration of naloxone, thiamine, and dextrose as indicated by the clinical picture. A focused history from bystanders or EMS and a targeted physical exam follow, guiding further investigations such as imaging or toxicology screening. This structured approach ensures immediately reversible causes are addressed before moving to more detailed workup.

#143 · Altered Mental Status and Behavioral Emergencies

What features distinguish delirium from dementia?

Quick Reference

Delirium has an acute onset over hours to days, a fluctuating course, and prominent inattention, often with altered level of consciousness. Dementia develops gradually over months to years with relatively stable attention, at least in earlier stages. A patient with dementia can still develop superimposed delirium, which should always be actively looked for.

#144 · Altered Mental Status and Behavioral Emergencies

What are the core components of the Confusion Assessment Method used to diagnose delirium?

Quick Reference

Acute onset with a fluctuating course, inattention, and either disorganized thinking or an altered level of consciousness. The diagnosis requires the first two features plus at least one of the latter two. This tool provides a structured, reproducible way to identify delirium at the bedside rather than relying on subjective impression alone.

What reversible causes should be actively sought in every patient presenting with delirium?

Quick Reference

Infection, medication effects or interactions, metabolic derangements such as electrolyte abnormalities or hypoglycemia, urinary retention, constipation, and uncontrolled pain are all common and correctable contributors. A thorough search for these causes is essential since delirium is a symptom of an underlying problem rather than a primary diagnosis. Treating the precipitant often resolves the delirium without needing prolonged sedative medication.

Why is hypoactive delirium often under-recognized compared to hyperactive delirium?

Quick Reference

Hypoactive delirium presents with lethargy, withdrawal, and reduced activity rather than the agitation and combativeness seen in hyperactive delirium, so it can be mistaken for depression or simple fatigue. Because it does not disrupt the unit or draw immediate attention, it is frequently missed on routine assessment. Deliberate screening with a tool like the CAM helps catch these quieter presentations.

#147 · Altered Mental Status and Behavioral Emergencies

What non-pharmacologic strategies are first-line for managing delirium?

Quick Reference

Reorientation with clocks, calendars, and familiar objects, encouraging family presence, minimizing unnecessary tethers like catheters and IV lines, and promoting normal sleep-wake cycles by reducing nighttime disruptions. Restraints should be avoided when possible since they can worsen agitation and increase injury risk. These measures are recommended as the foundation of management before turning to medications.

#148 · Altered Mental Status and Behavioral Emergencies

What is the preferred pharmacologic agent for agitated delirium when medication is needed, and what is generally avoided?

Quick Reference

Low-dose antipsychotics such as haloperidol are generally preferred for controlling dangerous agitation in delirium. Benzodiazepines are generally avoided in this setting because they can paradoxically worsen confusion in older or delirious patients, except when the delirium is due to alcohol or benzodiazepine withdrawal. Choosing the wrong class of medication can make the underlying confusion worse rather than better.

#149 · Altered Mental Status and Behavioral Emergencies

When is it appropriate to use benzodiazepines as first-line therapy in delirium?

Quick Reference

Benzodiazepines are first-line specifically for delirium caused by alcohol or sedative-hypnotic withdrawal, since they treat the underlying pathophysiology directly. Using them for delirium from other causes, such as infection or metabolic derangement, is generally avoided given the risk of worsening confusion. Recognizing the underlying cause is what determines which drug class is appropriate.

#150 · Altered Mental Status and Behavioral Emergencies

What is the typical timeline for delirium tremens after the last alcoholic drink, and how is it treated?

Quick Reference

Delirium tremens typically develops forty-eight to ninety-six hours after the last drink, following earlier withdrawal symptoms like tremor and anxiety. It is treated aggressively with benzodiazepines, often using a symptom-triggered protocol like CIWA-Ar to guide dosing. Untreated delirium tremens carries significant mortality, so early recognition and treatment of withdrawal is important.

#151 · Altered Mental Status and Behavioral Emergencies

What is excited delirium, and why does it require careful monitoring during sedation?

Quick Reference

It describes a state of severe agitation, often with stimulant intoxication or acute psychiatric decompensation, accompanied by extreme physical exertion and autonomic instability. Patients are at risk for sudden cardiac arrest from the combination of exertion, acidosis, and hyperthermia. Rapid sedation is often necessary, but must be paired with close cardiac and physiologic monitoring rather than physical restraint alone.

#152 · Altered Mental Status and Behavioral Emergencies

What findings distinguish anticholinergic toxicity as a cause of agitation and delirium?

Quick Reference

The classic description includes dry, flushed skin, dilated pupils, urinary retention, tachycardia, and hyperthermia, sometimes summarized informally with phrases like being hot, dry, red, and mad. This presentation should be distinguished from other causes of agitation since management focuses on supportive care and, in severe cases, physostigmine. Recognizing the specific pattern of findings helps identify the responsible toxin or medication.

#153 · Altered Mental Status and Behavioral Emergencies

How is serotonin syndrome distinguished from neuroleptic malignant syndrome in an agitated, hyperthermic patient?

Quick Reference

Serotonin syndrome typically has a rapid onset within hours of starting or increasing a serotonergic agent, with hyperreflexia, clonus, and tremor, especially in the lower extremities. Neuroleptic malignant syndrome develops more slowly over days after antipsychotic exposure, with bradyreflexia and characteristic lead-pipe muscle rigidity rather than clonus. Correctly distinguishing the two changes both the causative drug to stop and the specific treatment used.

#154 · Altered Mental Status and Behavioral Emergencies

What principles should guide the use of physical or chemical restraints in an agitated patient?

Quick Reference

The least restrictive intervention necessary should always be used first, with restraints reserved for situations where the patient poses a genuine risk to themselves or others. Clear documentation of the indication, ongoing reassessment, and legal or institutional policy requirements must all be followed. Restraints are a temporary safety measure, not a substitute for treating the underlying cause of agitation.

Why is a capacity assessment important in a delirious or agitated patient who wants to leave against medical advice?

Quick Reference

Delirium impairs the ability to understand information, appreciate consequences, and make a reasoned decision, meaning many delirious patients lack capacity to safely refuse care or discharge themselves. A formal capacity assessment should be documented before allowing such a patient to leave. Allowing an incapable patient to leave without addressing this can result in serious harm and legal liability.

Why are patients with underlying dementia more susceptible to delirium from a relatively minor illness?

Quick Reference

Reduced cognitive reserve means these patients have less capacity to compensate for physiologic stressors like a urinary tract infection, medication change, or mild dehydration, so even minor insults can precipitate a marked acute confusional state on top of their baseline impairment. This is sometimes described as a sudden worsening from baseline rather than a new diagnosis. Identifying and treating the precipitating illness typically improves the patient back toward their prior baseline.

#157 · Altered Mental Status and Behavioral Emergencies

Does directly asking a patient about suicidal thoughts increase their risk of acting on them?

Quick Reference

No, this is a persistent myth. Asking directly and specifically about suicidal ideation, plan, and intent does not increase risk and is a necessary part of a thorough assessment. Avoiding the question out of fear of planting the idea leads to missed risk and inadequate safety planning.

#158 · Altered Mental Status and Behavioral Emergencies

What is considered the single strongest risk factor for a future completed suicide?

Quick Reference

A prior suicide attempt is generally regarded as the strongest individual predictor of future risk. Other important risk factors include substance use, access to lethal means, social isolation, and a recent significant loss. A history of prior attempts should always prompt a more detailed and cautious current assessment.

What factors help protect against suicide risk and should be assessed alongside risk factors?

Quick Reference

Strong social support, religious or cultural beliefs that discourage suicide, a sense of responsibility toward children or dependents, and active engagement in mental health treatment are recognized protective factors. Assessing these alongside risk factors gives a more complete and balanced picture rather than focusing only on danger signs. The presence of strong protective factors can sometimes support a safer outpatient disposition.

What does the SAD PERSONS scale assess, and what is an important limitation of using it?

Quick Reference

It is a mnemonic-based checklist covering factors like sex, age, depression, prior attempt, ethanol use, rational thinking loss, social supports lacking, organized plan, no spouse, and sickness, used to help stratify suicide risk. Its main limitation is that it should support, not replace, careful individualized clinical judgment, since it was not designed as a stand-alone decision tool for disposition. Over-reliance on the numeric score alone can lead to inappropriate risk categorization.

Why is means restriction counseling an important part of suicide risk management?

Quick Reference

Reducing access to highly lethal means, such as firearms or large quantities of medication, has been shown to lower suicide risk, since many attempts occur impulsively and having immediate access to lethal means increases the chance of a fatal outcome. Counseling family members to secure or remove these items at home is a practical and evidence-based intervention. This is considered one of the most effective population-level suicide prevention strategies.

What factors are weighed when deciding whether a suicidal patient requires inpatient psychiatric hospitalization?

Quick Reference

The presence of a specific plan, clear intent, access to the means described in that plan, and the absence of adequate protective factors or social supports all push toward hospitalization. Passive ideation without a plan or intent, in a patient with good supports, may be manageable with close outpatient follow-up instead. The overall clinical picture, not any single factor alone, drives the disposition decision.

What is the difference between voluntary and involuntary psychiatric admission, and when is involuntary admission considered?

Quick Reference

Voluntary admission occurs when a patient agrees to hospitalization for their safety, while involuntary admission is used when a patient at significant risk refuses admission but meets the legal criteria for detention under local mental health legislation. Involuntary status requires meeting specific criteria, typically related to risk of harm to self or others due to a mental disorder. Clinicians must know their jurisdiction's specific legal criteria and process for involuntary admission.

How does passive suicidal ideation differ from active ideation with intent, and why does this distinction matter for management?

Quick Reference

Passive ideation involves thoughts of wanting to be dead or that life is not worth living without a specific plan or intent to act, while active ideation with intent involves a clear wish and readiness to act, sometimes with a defined plan. Active ideation with intent and a plan generally requires a much more urgent and restrictive response, potentially including hospitalization. Treating all suicidal ideation identically without distinguishing severity can lead to either under- or over-triage.

Why are no-suicide contracts no longer considered adequate on their own for managing suicide risk?

Quick Reference

Evidence does not support that a patient's verbal or written promise not to harm themselves meaningfully reduces actual risk, and relying on such a contract can create false reassurance for the treating clinician. A thorough, structured risk assessment and an individualized safety plan are more appropriate than a simple promise. Documentation should reflect the actual risk assessment performed, not just that a contract was signed.

What elements should be included in discharge planning for a patient assessed as safe for outpatient management after a suicide risk assessment?

Quick Reference

A concrete written safety plan, confirmed follow-up with mental health services, involvement of family or other supports where appropriate, and specific means restriction counseling tailored to the patient's situation. Vague instructions to follow up with a family doctor without a concrete plan are inadequate. Clear communication with the patient about what to do if symptoms worsen before the follow-up appointment is also essential.

Why must medical stabilization take priority over psychiatric assessment immediately after a suicide attempt?

Quick Reference

Life-threatening medical issues from the attempt itself, such as toxic ingestion, hemorrhage, or hypoxia, must be addressed first since an unstable patient cannot meaningfully participate in a psychiatric interview and delaying treatment risks death. Once medically stable, a thorough psychiatric risk assessment should follow promptly. Skipping or rushing medical stabilization to get to the psychiatric assessment sooner is a dangerous sequencing error.

What unique risk factors should be considered when assessing suicide risk in adolescents?

Quick Reference

Bullying, including cyberbullying, contagion effects after exposure to a peer's suicide, and increased risk among LGBTQ+ youth facing rejection or discrimination are important adolescent-specific considerations. Family dynamics and school-related stressors also play a larger role in this age group compared to adults. A risk assessment tailored to these factors gives a more accurate picture than applying adult-focused criteria alone.

#169 · Altered Mental Status and Behavioral Emergencies

Why is suicide risk in elderly patients often considered particularly dangerous despite lower rates of suicidal ideation being reported?

Quick Reference

Elderly patients who do attempt suicide tend to use more immediately lethal means and are less likely to survive an attempt or be discovered in time, giving this age group one of the highest suicide completion rates of any population. Depression in the elderly is also frequently underdiagnosed and undertreated, contributing to unaddressed risk. Any expression of hopelessness or suicidal thinking in an older patient should be taken very seriously.

#170 · Altered Mental Status and Behavioral Emergencies

Why is it a mistake to dismiss a frequent emergency department presenter with suicidal statements as simply malingering?

Quick Reference

Frequent presentations can reflect genuine chronic risk, an unmet need for mental health support, or a worsening underlying condition, and assuming malingering without a full reassessment each visit can lead to a missed true crisis. Each presentation deserves its own individualized risk assessment rather than relying on the patient's prior pattern of visits. Fatigue or frustration with frequent presenters should never substitute for careful clinical evaluation.

#171 · Trauma and Musculoskeletal Injury

What is the correct order of priorities in the primary survey for a multi-trauma patient?

Quick Reference

Airway with cervical spine protection, Breathing, Circulation with hemorrhage control, Disability (neuro status), Exposure with environmental control. This ABCDE sequence never changes, and each letter must be addressed before moving to the next even if a later problem looks more dramatic.

#172 · Trauma and Musculoskeletal Injury

A trauma patient has a rigid, distended abdomen and hypotension unresponsive to two litres of crystalloid. What is the next step?

Quick Reference

Activate massive transfusion and arrange emergent surgery or interventional radiology rather than giving more crystalloid. Ongoing hemodynamic instability with a tender abdomen after trauma means intra-abdominal hemorrhage until proven otherwise, and repeated fluid boluses only delay definitive hemorrhage control.

#173 · Trauma and Musculoskeletal Injury

What clinical triad suggests a tension pneumothorax before imaging is available?

Quick Reference

Absent breath sounds on one side, tracheal deviation away from that side, and distended neck veins with hypotension. This is a clinical diagnosis requiring immediate needle or finger thoracostomy, not a chest X-ray, because waiting for imaging can be fatal.

#174 · Trauma and Musculoskeletal Injury

Why is a blind nasogastric tube contraindicated in some trauma patients?

Quick Reference

In suspected basilar skull fracture (raccoon eyes, Battle sign, CSF rhinorrhea or otorrhea) a nasogastric tube can pass through the fracture into the cranial vault. An orogastric tube should be used instead when these signs are present.

#175 · Trauma and Musculoskeletal Injury

What is the FAST exam used for in trauma, and what is its main limitation?

Quick Reference

Focused Assessment with Sonography for Trauma looks for free fluid in the pericardial sac, hepatorenal space, splenorenal space, and pelvis, suggesting hemorrhage. A negative FAST does not rule out significant injury, especially retroperitoneal or hollow viscus injury, so clinical reassessment is still required.

#176 · Trauma and Musculoskeletal Injury

What does a Glasgow Coma Scale of 8 or less mean for airway management?

Quick Reference

GCS 8 or less generally means the patient cannot protect their own airway and needs definitive airway management, usually endotracheal intubation. This threshold is a classic exam trigger point for airway decisions in trauma.

#177 · Trauma and Musculoskeletal Injury

A candidate orders a pelvic X-ray last in a hypotensive blunt trauma patient. Why is this a trap?

Quick Reference

Pelvic fracture is a major source of occult hemorrhage and should be assessed early, often with a pelvic binder applied empirically in unstable patients before imaging confirms the fracture. Delaying pelvic stabilization allows ongoing venous and bony bleeding.

#178 · Trauma and Musculoskeletal Injury

What is permissive hypotension in trauma resuscitation?

Quick Reference

In penetrating trauma with uncontrolled hemorrhage, fluid resuscitation is deliberately limited to keep systolic pressure around 80 to 90 mmHg until bleeding is surgically controlled. Normalizing blood pressure too early can dislodge clots and worsen bleeding.

#179 · Trauma and Musculoskeletal Injury

What are the indications to log-roll a trauma patient and what must be checked?

Quick Reference

The log roll is performed to inspect and palpate the entire back, spine, and buttocks for wounds, step-offs, or tenderness while maintaining spinal precautions. It is also the moment to check for a rectal tone and blood if indicated.

#180 · Trauma and Musculoskeletal Injury

A patient with a stab wound to the left lower chest is stable. What organ injury must be considered even though the wound looks thoracic?

Quick Reference

The diaphragm can rise as high as the fourth intercostal space on expiration, so any penetrating wound below the nipple line or scapula tip can injure abdominal organs including spleen, stomach, or diaphragm. This is a classic trap where a chest wound is treated as purely thoracic.

#181 · Trauma and Musculoskeletal Injury

What does the mnemonic AMPLE stand for in trauma history taking?

Quick Reference

Allergies, Medications, Past medical history, Last meal, and Events/Environment surrounding the injury. This structured history is gathered efficiently during or right after the primary and secondary surveys.

#182 · Trauma and Musculoskeletal Injury

Why is hypothermia dangerous in a bleeding trauma patient, and what is it part of?

Quick Reference

Hypothermia worsens coagulopathy and platelet dysfunction, forming part of the lethal triad with acidosis and coagulopathy that accelerates death from hemorrhage. Active warming with blankets and warmed fluids is part of standard trauma resuscitation.

#183 · Trauma and Musculoskeletal Injury

A trauma patient has jugular venous distension, muffled heart sounds, and hypotension. What is the diagnosis and immediate management?

Quick Reference

This is Beck triad, suggesting cardiac tamponade from pericardial blood. Immediate management is pericardiocentesis or emergent thoracotomy if the patient is peri-arrest, with definitive repair in the operating room.

#184 · Trauma and Musculoskeletal Injury

What is the significance of a seatbelt sign across the abdomen in a motor vehicle collision?

Quick Reference

A seatbelt sign is associated with an increased risk of intestinal and mesenteric injury, including bowel perforation, even when initial imaging looks unremarkable. These patients need close serial abdominal examination and a low threshold for repeat imaging or surgical consultation.

#185 · Trauma and Musculoskeletal Injury

When should the secondary survey begin in trauma assessment?

Quick Reference

The secondary survey, a head-to-toe examination looking for all injuries, begins only after the primary survey is complete and any immediately life-threatening problems in airway, breathing, or circulation have been addressed. Starting it too early risks missing a rapidly deteriorating ABC problem.

#186 · Trauma and Musculoskeletal Injury

A patient falls on an outstretched hand and has snuffbox tenderness with a normal X-ray. What should be done?

Quick Reference

Scaphoid fractures are frequently occult on initial X-ray, so the wrist should be immobilized in a thumb spica splint and repeat imaging or advanced imaging arranged in 10 to 14 days. Missing a scaphoid fracture risks avascular necrosis because of its retrograde blood supply.

#187 · Trauma and Musculoskeletal Injury

What are the five Ps of compartment syndrome, and which is most reliable early?

Quick Reference

Pain out of proportion, Pallor, Paresthesia, Pulselessness, and Paralysis. Pain out of proportion to the injury and pain with passive stretch of the muscles is the earliest and most reliable sign, since pulselessness is a late finding that means the diagnosis was missed earlier.

#188 · Trauma and Musculoskeletal Injury

Why must an open fracture be treated differently than a closed fracture in the emergency department?

Quick Reference

Open fractures need early intravenous antibiotics, tetanus prophylaxis assessment, and urgent orthopedic consultation for operative irrigation and debridement to reduce infection risk. Time to antibiotics and definitive washout are key quality measures, not just splinting the limb.

#189 · Trauma and Musculoskeletal Injury

A posterior hip dislocation classically presents with the leg in what position, and what nerve is at risk?

Quick Reference

The leg is shortened, internally rotated, and adducted. The sciatic nerve is at risk of injury and should be examined before and after any reduction attempt, along with distal pulses.

#190 · Trauma and Musculoskeletal Injury

What is a Monteggia fracture and why is it commonly missed?

Quick Reference

A Monteggia fracture is an ulnar shaft fracture with dislocation of the radial head. It is commonly missed when the elbow joint is not adequately visualized on X-ray, so any forearm fracture requires imaging of the joint above and below.

#191 · Trauma and Musculoskeletal Injury

A young athlete has anterior knee pain and a positive anterior drawer test after a pivoting injury. What structure is likely torn?

Quick Reference

This presentation suggests an anterior cruciate ligament tear, often accompanied by an audible pop at the time of injury and rapid knee swelling from hemarthrosis. MRI confirms the diagnosis but management in the emergency department is analgesia, immobilization, and orthopedic or sports medicine follow-up.

#192 · Trauma and Musculoskeletal Injury

What are the Ottawa Ankle Rules used for and what do they include?

Quick Reference

They help decide whether an ankle X-ray is needed after acute injury, reducing unnecessary imaging. Imaging is indicated if there is bone tenderness at the posterior edge or tip of either malleolus, or inability to bear weight for four steps both immediately and in the emergency department.

#193 · Trauma and Musculoskeletal Injury

A supracondylar humerus fracture in a child requires assessment of which specific nerve and vessel?

Quick Reference

The anterior interosseous branch of the median nerve and the brachial artery are at highest risk, so the exam should specifically check the OK sign for anterior interosseous function and check the radial pulse and capillary refill. Vascular compromise in this fracture is a limb-threatening emergency.

#194 · Trauma and Musculoskeletal Injury

Why should a candidate avoid describing every long bone fracture the same way?

Quick Reference

Fracture description should include location, pattern (transverse, oblique, spiral, comminuted), displacement, angulation, and whether it is open or closed, since management and disposition depend on these details, not just the bone involved. A spiral fracture in a young child, for instance, raises concern for non-accidental injury.

#195 · Trauma and Musculoskeletal Injury

What is the immediate management priority for a fracture with obvious neurovascular compromise before transport to imaging?

Quick Reference

Gentle reduction or realignment should be attempted at the bedside to restore perfusion, rather than sending the limb to radiology pulseless. Documenting neurovascular status before and after any manipulation is essential.

#196 · Trauma and Musculoskeletal Injury

A patient has a clavicle fracture in the middle third with skin tenting. What is the concern?

Quick Reference

Skin tenting over a displaced clavicle fracture threatens to become an open fracture if the skin breaks down, and it can also signal impending compromise of nearby neurovascular structures. Orthopedic consultation is warranted rather than routine sling and follow-up alone.

#197 · Trauma and Musculoskeletal Injury

What historical or exam feature should raise suspicion for a pathologic fracture rather than a purely traumatic one?

Quick Reference

A fracture occurring with minimal or no trauma, especially in an older adult or a patient with known malignancy, osteoporosis, or bone lesions, suggests pathologic fracture. This should prompt evaluation for an underlying bone lesion rather than treating it as a simple mechanical fracture.

#198 · Trauma and Musculoskeletal Injury

A boxer's fracture involves which bone and what is an acceptable angulation before it needs orthopedic reduction?

Quick Reference

It is a fracture of the fifth metacarpal neck, typically from punching. Angulation up to about 40 to 45 degrees is often accepted with buddy taping or an ulnar gutter splint, but rotational deformity of the finger, even a small amount, is not acceptable and needs reduction.

#199 · Trauma and Musculoskeletal Injury

Why is it a trap to reassure a patient with a normal X-ray after a twisting knee injury and ongoing effusion?

Quick Reference

A normal X-ray does not exclude significant ligamentous or meniscal injury, since these soft tissue structures are not well seen on plain films. Persistent effusion, instability, or locking symptoms should prompt further assessment rather than discharge with reassurance alone.

#200 · Trauma and Musculoskeletal Injury

What is the general time window after which a simple laceration is usually still considered safe for primary closure?

Quick Reference

Most simple lacerations can be closed primarily up to about 6 to 8 hours after injury, though highly vascular areas like the face can often be closed later. Beyond this window, and in heavily contaminated wounds, the infection risk of primary closure rises and delayed closure may be preferred.

#201 · Trauma and Musculoskeletal Injury

Before closing any hand laceration, what specific functions must be tested and documented?

Quick Reference

Motor function of flexor and extensor tendons, sensation in each nerve distribution, and vascular status distal to the wound must all be checked before local anesthesia is given. Testing after anesthesia is too late because pain and altered sensation can mask a real deficit.

#202 · Trauma and Musculoskeletal Injury

A patient has a laceration over a joint with clear or straw-colored fluid draining from the wound. What must be ruled out?

Quick Reference

This raises concern for a joint capsule injury with synovial fluid leak, sometimes tested by injecting saline into the joint to see if it exits the wound. Missing a joint space injury risks septic arthritis if it is closed without recognition.

#203 · Trauma and Musculoskeletal Injury

Why are cat bite wounds treated more aggressively with prophylactic antibiotics than most dog bites?

Quick Reference

Cat teeth are sharp and narrow, driving bacteria like *Pasteurella multocida* deep into tissue and joints while leaving a small entry wound, giving a much higher infection rate than dog bites. Prophylactic antibiotics are recommended for most cat bites, especially on the hand.

#204 · Trauma and Musculoskeletal Injury

What is the correct approach to a wound with a suspected retained foreign body that is not visible on inspection?

Quick Reference

Radiopaque foreign bodies like glass or metal are seen on plain X-ray, but organic material such as wood is often missed and may need ultrasound. Any wound from a mechanism suggesting a foreign body should be imaged rather than assumed clean based on visual inspection alone.

#205 · Trauma and Musculoskeletal Injury

When is tetanus immunoglobulin indicated in addition to a tetanus toxoid booster for a wound?

Quick Reference

Tetanus immunoglobulin is given for tetanus-prone wounds (dirty, contaminated, or with devitalized tissue) in patients whose immunization history is unknown or incomplete, meaning fewer than three prior doses. A clean minor wound in a fully immunized patient needs neither immunoglobulin nor an early booster.

#206 · Trauma and Musculoskeletal Injury

What laceration feature makes primary closure inappropriate and favors healing by secondary intention or delayed closure?

Quick Reference

Heavily contaminated wounds, established infection, significant tissue loss, or wounds more than the safe time window old are generally left open or closed later rather than closed primarily. Closing a contaminated wound tightly traps bacteria and increases the risk of abscess formation.

#207 · Trauma and Musculoskeletal Injury

A candidate uses lidocaine with epinephrine on a finger laceration. Is this acceptable practice?

Quick Reference

Current evidence supports that epinephrine-containing local anesthetic is safe in digits with normal vascular supply, contrary to the older teaching that it always causes digital ischemia. It should still be avoided in patients with known peripheral vascular disease or compromised digital circulation.

#208 · Trauma and Musculoskeletal Injury

What is the most important step before repairing any traumatic wound, regardless of size?

Quick Reference

Thorough irrigation with a large volume of normal saline or tap water under adequate pressure to remove debris and reduce bacterial load is the single most important step in preventing wound infection. Suturing without adequate irrigation is a common and preventable error.

#209 · Trauma and Musculoskeletal Injury

A facial laceration crosses the vermilion border of the lip. What is critical to get right during repair?

Quick Reference

The vermilion border must be aligned precisely as the very first suture, before any other part of the laceration is closed, because even a 1 to 2 millimetre step-off is cosmetically obvious for life. This is a classic scenario testing attention to landmark alignment.

#210 · Trauma and Musculoskeletal Injury

What wound characteristic makes tissue adhesive (skin glue) an inappropriate closure choice?

Quick Reference

Tissue adhesive should be avoided over high-tension areas, joints, mucosal surfaces, and wounds requiring deep sutures, since it only approximates the epidermis and cannot hold tension or close dead space. It works best on low-tension, clean, linear lacerations.

#211 · Trauma and Musculoskeletal Injury

A patient returns 3 days after a hand laceration repair with increasing redness, swelling, and pain around the wound. What is the next step?

Quick Reference

This suggests early wound infection and requires removing some sutures to allow drainage, wound culture if purulent, and starting appropriate antibiotics rather than simply reassuring the patient. Ignoring a spreading infection risks progression to abscess or deeper tissue involvement.

#212 · Trauma and Musculoskeletal Injury

Why should a scalp laceration in a young child prompt a careful neurologic and mechanism history?

Quick Reference

Scalp wounds can bleed profusely and mask an underlying skull fracture or intracranial injury, so the mechanism, any loss of consciousness, and neurologic exam must be assessed even if the wound itself looks straightforward. This is a common trap of focusing only on the visible laceration.

#213 · Trauma and Musculoskeletal Injury

What generally determines whether a laceration should be repaired with sutures, staples, or adhesive strips?

Quick Reference

Location, tension across the wound, and cosmetic priority guide the choice: high-tension or mobile areas usually need sutures, hair-bearing scalp areas often do well with staples, and low-tension areas can use adhesive strips or glue. Choosing based on convenience alone rather than wound characteristics increases dehiscence risk.

#214 · Trauma and Musculoskeletal Injury

How is the rule of nines used to estimate total body surface area burned in an adult?

Quick Reference

Each arm is about 9 percent, each leg about 18 percent, the anterior and posterior trunk are each about 18 percent, and the head is about 9 percent, with the palm of the patient's own hand representing roughly 1 percent for smaller irregular burns. This estimate guides fluid resuscitation and transfer decisions.

#215 · Trauma and Musculoskeletal Injury

What formula is used to calculate fluid resuscitation in major burns, and when does the clock start?

Quick Reference

The Parkland formula gives 4 mL of crystalloid per kilogram per percent body surface area burned over 24 hours, with half given in the first 8 hours. The clock starts at the time of injury, not the time the patient arrives in the emergency department, which is a common calculation trap.

#216 · Trauma and Musculoskeletal Injury

A patient has facial burns, singed nasal hair, and a hoarse voice after a house fire. What is the priority concern?

Quick Reference

These findings suggest inhalation injury with impending airway edema, and early elective intubation should be strongly considered before swelling makes the airway impossible to secure. Waiting until stridor develops is often too late.

#217 · Trauma and Musculoskeletal Injury

How do superficial, partial-thickness, and full-thickness burns differ in appearance and sensation?

Quick Reference

Superficial burns are red and painful without blisters, partial-thickness burns are blistered, moist, and very painful, and full-thickness burns are dry, leathery, and often painless because nerve endings are destroyed. A painless burn is a trap that can be mistaken for a minor injury when it is actually the most severe.

#218 · Trauma and Musculoskeletal Injury

What carbon monoxide poisoning finding can falsely reassure a clinician relying on standard monitoring?

Quick Reference

Pulse oximetry cannot distinguish oxyhemoglobin from carboxyhemoglobin and will read falsely normal or high in carbon monoxide poisoning. A co-oximeter or blood gas measuring carboxyhemoglobin level directly is needed when inhalation injury is suspected.

#219 · Trauma and Musculoskeletal Injury

A circumferential full-thickness burn to a limb causes worsening pain, pallor, and diminished pulses. What procedure is indicated?

Quick Reference

Escharotomy, a surgical incision through the non-elastic burned skin, is needed to relieve pressure and restore perfusion, since the eschar itself acts like a tourniquet as tissue swells underneath it. This is distinct from fasciotomy, which goes deeper into the muscle compartments.

#220 · Trauma and Musculoskeletal Injury

What burn size and depth criteria generally mandate transfer to a specialized burn centre?

Quick Reference

Partial-thickness burns over more than 10 percent body surface area, any full-thickness burn, burns involving the face, hands, feet, genitalia, or major joints, and inhalation or electrical burns generally warrant burn centre transfer. Age extremes and comorbidities lower the threshold further.

#221 · Trauma and Musculoskeletal Injury

Why should ice never be applied directly to a burn wound?

Quick Reference

Ice or very cold water can cause vasoconstriction and worsen tissue injury, effectively converting a partial-thickness burn to a deeper one, and can contribute to hypothermia in larger burns. Cool (not ice-cold) running water for a limited time is the appropriate initial first aid.

#222 · Trauma and Musculoskeletal Injury

A burn pattern shows sharply demarcated stocking or glove distribution with sparing of flexor creases in a toddler. What should be considered?

Quick Reference

This pattern is classic for an intentional immersion burn (non-accidental injury), since accidental splash burns typically have irregular edges and splash marks rather than sharp uniform lines. Any suspicious burn pattern in a child requires further investigation and appropriate reporting.

#223 · Trauma and Musculoskeletal Injury

What distinguishes management of a chemical burn from a thermal burn in initial first aid?

Quick Reference

Chemical burns require prolonged irrigation, often 20 minutes or more, with copious water to dilute and remove the agent, and dry powder chemicals should be brushed off before irrigation begins. Neutralizing agents are generally avoided because the reaction can generate additional heat and worsen injury.

#224 · Trauma and Musculoskeletal Injury

A patient has a high-voltage electrical burn with small entry and exit wounds but appears otherwise well. What internal injury pattern should still be suspected?

Quick Reference

Electrical current can cause extensive deep tissue and muscle damage along its internal path that is disproportionate to the small visible skin wounds, including myoglobinuria from rhabdomyolysis and cardiac arrhythmia. Cardiac monitoring, creatine kinase, and urine myoglobin should be checked even when the skin looks reassuring.

#225 · Trauma and Musculoskeletal Injury

Why is early analgesia important in burn management beyond patient comfort?

Quick Reference

Burns are extremely painful and undertreated pain can worsen physiologic stress response and complicate wound assessment and dressing changes. Adequate parenteral analgesia should be given promptly rather than delayed until after all assessments are complete.

#226 · Trauma and Musculoskeletal Injury

What is a key trap when estimating burn size in a patient with scattered small burns rather than one large area?

Quick Reference

Using the patient's own palm (including fingers) as approximately 1 percent body surface area is more accurate for scattered small or irregular burns than trying to apply the rule of nines, which is designed for larger contiguous areas. Overestimating scattered burns can lead to unnecessary aggressive fluid resuscitation.

#227 · Trauma and Musculoskeletal Injury

Should burn blisters generally be deroofed or left intact in initial emergency management?

Quick Reference

Practice varies, but many guidelines favor leaving small intact blisters alone since the blister roof protects the wound bed, while large, tense, or likely-to-rupture blisters may be deroofed to allow proper wound assessment and dressing. The key exam point is that automatically popping every blister is not correct practice.

#228 · Abdominal and Gastrointestinal Emergencies

A postmenopausal patient with abdominal pain has left lower quadrant tenderness and fever. What is the leading diagnosis and what must always be excluded first in a woman of reproductive age with similar pain?

Quick Reference

In an older patient this points to diverticulitis, but in any woman of reproductive age with lower abdominal pain, ectopic pregnancy must be excluded with a pregnancy test before other differentials are pursued. Missing pregnancy status is one of the most dangerous errors in abdominal pain assessment.

#229 · Abdominal and Gastrointestinal Emergencies

What is the classic migration pattern of pain in appendicitis, and why can it be atypical in pregnancy?

Quick Reference

Classically pain starts periumbilically and migrates to the right lower quadrant (McBurney point) as inflammation progresses to involve the parietal peritoneum. In pregnancy, the enlarging uterus displaces the appendix superiorly and laterally, so pain may present in the right upper quadrant or flank instead.

#230 · Abdominal and Gastrointestinal Emergencies

A patient with right upper quadrant pain, fever, and jaundice has what classic triad, and what does it suggest?

Quick Reference

This is Charcot triad, suggesting ascending cholangitis from biliary obstruction, typically due to a common bile duct stone. This is a surgical and often septic emergency requiring urgent biliary decompression, not simple outpatient antibiotics.

#231 · Abdominal and Gastrointestinal Emergencies

An older patient presents with sudden severe abdominal pain out of proportion to a benign-appearing exam. What diagnosis must be considered?

Quick Reference

Pain out of proportion to physical findings is the classic description of acute mesenteric ischemia, especially in patients with atrial fibrillation or vascular disease. Waiting for peritonitis to develop before considering this diagnosis delays care of a condition with high mortality.

#232 · Abdominal and Gastrointestinal Emergencies

What abdominal pain presentation with hypotension should immediately raise concern for a ruptured abdominal aortic aneurysm?

Quick Reference

Sudden severe back or flank pain with hypotension, especially in an older patient with a pulsatile abdominal mass, should trigger immediate bedside ultrasound and vascular surgery consultation rather than routine workup. This diagnosis can be rapidly fatal if not recognized promptly.

#233 · Abdominal and Gastrointestinal Emergencies

Why can elderly patients present atypically with intra-abdominal catastrophes like perforation or ischemia?

Quick Reference

Older adults often have blunted pain response, less pronounced peritoneal signs, and may present with vague complaints, confusion, or minimal fever despite serious underlying pathology. A benign-appearing exam in an elderly patient should not be reassuring without a lower threshold for imaging.

#234 · Abdominal and Gastrointestinal Emergencies

A patient with epigastric pain radiating to the back has markedly elevated lipase. What complication should be assessed for at presentation?

Quick Reference

Acute pancreatitis can rapidly cause third-spacing, hypovolemia, and systemic inflammatory response, so early aggressive intravenous fluid resuscitation and monitoring for signs of severity (such as hypotension, hypoxia, or organ dysfunction) are essential from the start, not just pain control.

#235 · Abdominal and Gastrointestinal Emergencies

What historical clue on abdominal pain assessment should prompt consideration of small bowel obstruction?

Quick Reference

Colicky, crampy abdominal pain with vomiting, distension, and obstipation (no passage of flatus or stool), particularly in a patient with prior abdominal surgery, suggests small bowel obstruction from adhesions. Prior surgical history is a key detail that is easy to overlook if not specifically asked about.

#236 · Abdominal and Gastrointestinal Emergencies

A young woman has sudden severe unilateral lower abdominal pain with a known ovarian cyst. What time-sensitive diagnosis must be considered?

Quick Reference

Ovarian torsion should be considered, and pelvic ultrasound with Doppler flow should be obtained urgently since ovarian salvage depends on rapid diagnosis and surgical detorsion. Absent flow on Doppler does not fully exclude torsion, so clinical suspicion should still drive urgent gynecology consultation.

#237 · Abdominal and Gastrointestinal Emergencies

What is a common trap when a patient's abdominal pain temporarily improves after perforation of a hollow viscus?

Quick Reference

Sudden improvement in pain after a period of severe pain can reflect perforation with transient relief of pressure, not resolution of the underlying problem, and is often followed by worsening peritonitis as contents spread. This should not be mistaken for clinical improvement.

#238 · Abdominal and Gastrointestinal Emergencies

Why should a urinalysis alone not be relied upon to exclude a serious cause of flank pain in an older patient?

Quick Reference

Hematuria can occur with both renal colic and a leaking abdominal aortic aneurysm, so an older patient with flank pain and risk factors for vascular disease needs imaging to exclude aneurysm even if the urinalysis supports a stone. Assuming renal colic based on urinalysis alone in this population is a dangerous shortcut.

#239 · Abdominal and Gastrointestinal Emergencies

A patient with diffuse abdominal pain has a rigid, board-like abdomen with absent bowel sounds. What does this exam finding suggest and what is the disposition?

Quick Reference

This suggests diffuse peritonitis, often from perforation, and warrants immediate surgical consultation and preparation for the operating room rather than prolonged emergency department workup. Involuntary guarding and rigidity are the exam findings that should not be missed or delayed on.

#240 · Abdominal and Gastrointestinal Emergencies

What imaging modality is preferred first-line for suspected appendicitis in a pregnant patient or child, and why?

Quick Reference

Ultrasound is preferred first-line to avoid ionizing radiation in pregnancy and in children, reserving computed tomography for cases where ultrasound is inconclusive and the diagnosis remains uncertain. Reflexively ordering CT without considering radiation exposure is a common exam trap.

#241 · Abdominal and Gastrointestinal Emergencies

A patient reports abdominal pain that is severe overnight but the physical exam and initial bloodwork look reassuring in the morning. What approach best manages diagnostic uncertainty?

Quick Reference

Serial abdominal examinations over a period of observation, with reassessment of vital signs and repeat exam, are more valuable than relying on a single normal snapshot, since early presentations of surgical abdomens can look deceptively benign. Discharge without any reassessment plan risks missing an evolving process.

#242 · Abdominal and Gastrointestinal Emergencies

In right upper quadrant pain, what sign supports acute cholecystitis on physical exam?

Quick Reference

A positive Murphy sign, inspiratory arrest during deep palpation under the right costal margin due to pain, supports acute cholecystitis. It should prompt right upper quadrant ultrasound looking for gallstones, wall thickening, and pericholecystic fluid.

#243 · Abdominal and Gastrointestinal Emergencies

What is the first priority in managing a patient with a major upper gastrointestinal bleed, before diagnosis of the exact source?

Quick Reference

Hemodynamic resuscitation with large-bore intravenous access, crystalloid, and blood products as needed comes first, following the same ABC approach as any hemorrhage. Endoscopy is important for diagnosis and treatment but should not delay stabilization of an unstable patient.

#244 · Abdominal and Gastrointestinal Emergencies

How can melena versus hematochezia help localize a gastrointestinal bleed, and what is the exception to remember?

Quick Reference

Melena (black, tarry stool) usually suggests an upper source, while hematochezia (bright red blood per rectum) usually suggests a lower source. However, a brisk upper gastrointestinal bleed can also present as hematochezia due to rapid transit, so hematochezia does not always exclude an upper source in an unstable patient.

#245 · Abdominal and Gastrointestinal Emergencies

A patient with known cirrhosis presents with hematemesis. What specific medication class should be started early, in addition to resuscitation?

Quick Reference

Octreotide or another somatostatin analogue should be started early to reduce splanchnic blood flow and portal pressure in suspected variceal bleeding, alongside prophylactic antibiotics, since variceal hemorrhage in cirrhotic patients carries high mortality. Endoscopy is still needed but medical therapy should not wait for it.

#246 · Abdominal and Gastrointestinal Emergencies

What scoring system helps risk-stratify upper gastrointestinal bleeds and can support safe discharge in low-risk patients?

Quick Reference

The Glasgow-Blatchford score uses clinical and laboratory variables (without requiring endoscopy) to identify patients at very low risk who may be safely managed as outpatients with urgent follow-up, versus higher-risk patients needing admission and urgent endoscopy. Relying on hemoglobin alone underestimates risk in acute bleeding.

#247 · Abdominal and Gastrointestinal Emergencies

Why can hemoglobin be falsely normal early in an acute significant gastrointestinal bleed?

Quick Reference

Hemoglobin reflects the concentration of red cells in plasma, and in acute hemorrhage both red cells and plasma are lost proportionally at first, so hemoglobin may not drop until hemodilution occurs over hours as fluid shifts. A single normal hemoglobin at triage should not be falsely reassuring in a patient with signs of significant bleeding.

#248 · Abdominal and Gastrointestinal Emergencies

A patient on warfarin presents with a significant gastrointestinal bleed. What is an appropriate reversal approach?

Quick Reference

Prothrombin complex concentrate provides rapid reversal of vitamin K antagonist effect and is preferred over fresh frozen plasma alone in significant bleeding because of faster correction and smaller volume, with vitamin K given as well for sustained reversal. Waiting for fresh frozen plasma alone delays correction in an actively bleeding patient.

#249 · Abdominal and Gastrointestinal Emergencies

What historical feature should raise concern for an aortoenteric fistula in a patient with a gastrointestinal bleed?

Quick Reference

A history of prior abdominal aortic aneurysm repair with a graft, presenting with a herald bleed (a smaller bleed followed by a delay before a massive bleed), should raise concern for aortoenteric fistula. This rare but catastrophic diagnosis is easily missed if the surgical history is not specifically asked about.

#250 · Abdominal and Gastrointestinal Emergencies

What is the significance of orthostatic vital sign changes in a patient with suspected gastrointestinal bleeding and normal supine vitals?

Quick Reference

A significant rise in heart rate or drop in blood pressure with standing can indicate clinically significant volume loss even when supine vitals appear normal, suggesting a larger bleed than initially apparent. This should raise the acuity of management rather than being dismissed as a benign finding.

#251 · Abdominal and Gastrointestinal Emergencies

Why is nasogastric aspiration of limited value in ruling out an upper gastrointestinal bleed?

Quick Reference

A negative or non-bloody nasogastric aspirate does not reliably exclude an upper gastrointestinal source, since bleeding may be intermittent or arise beyond the pylorus with a competent pyloric valve preventing reflux into the stomach. It should not be used alone to decide a bleed is not from an upper source.

#252 · Abdominal and Gastrointestinal Emergencies

A patient with a lower gastrointestinal bleed and hemodynamic instability needs localization. What is generally the preferred initial investigation once stabilized?

Quick Reference

Colonoscopy after bowel preparation is often preferred once the patient is stabilized, though in ongoing brisk bleeding or an unprepped unstable patient, computed tomography angiography or nuclear medicine bleeding scan may be used to localize the source first. The choice depends on stability and bleeding rate, not a single fixed test for every patient.

#253 · Abdominal and Gastrointestinal Emergencies

What is a Mallory-Weiss tear and what history commonly precedes it?

Quick Reference

A Mallory-Weiss tear is a mucosal laceration at the gastroesophageal junction, classically following forceful vomiting or retching, often after heavy alcohol intake, that leads to hematemesis. Most cases are self-limited, but ongoing bleeding still requires endoscopic evaluation.

#254 · Abdominal and Gastrointestinal Emergencies

In a patient taking NSAIDs who presents with an upper gastrointestinal bleed, what medication should be started promptly in the emergency department?

Quick Reference

A proton pump inhibitor should be started promptly, as high-dose therapy before endoscopy can reduce the likelihood of high-risk stigmata found on endoscopy and reduce rebleeding risk. Waiting until after endoscopy to start acid suppression delays a beneficial intervention.

#255 · Abdominal and Gastrointestinal Emergencies

Why is it a trap to reassure a patient with a small amount of visible blood in vomit or stool without asking about anticoagulant or antiplatelet use?

Quick Reference

Patients on anticoagulants, antiplatelets, or with underlying coagulopathy can progress rapidly from a small visible bleed to a life-threatening one, so medication history changes both risk stratification and immediate management (including reversal strategy). A small visible amount of blood does not predict the true severity of the underlying lesion.

#256 · Abdominal and Gastrointestinal Emergencies

What is the typical target hemoglobin threshold for transfusion in most stable gastrointestinal bleeding patients, and why not transfuse more liberally?

Quick Reference

A restrictive transfusion strategy targeting a hemoglobin around 70 g/L (higher in patients with active cardiac ischemia) is generally preferred over liberal transfusion, since over-transfusion in variceal bleeding especially can raise portal pressure and increase rebleeding risk. Transfusing to a normal hemoglobin in every case is not correct practice.

#257 · Toxicology, Environmental Emergencies, and Anaphylaxis

What is the antidote for acetaminophen overdose and what determines whether it is given?

Quick Reference

N-acetylcysteine is the antidote, and the decision to treat is guided by the acetaminophen concentration plotted on the Rumack-Matthew nomogram against time since a single acute ingestion. Treatment should not be delayed while awaiting a level if the history suggests a large ingestion or if timing is unclear, since early treatment is most effective.

#258 · Toxicology, Environmental Emergencies, and Anaphylaxis

A patient presents with an anion gap metabolic acidosis, altered mental status, and visual disturbances after a witnessed ingestion. What toxin should be considered and what is the antidote?

Quick Reference

Methanol poisoning should be considered, causing an elevated osmolar and anion gap along with visual symptoms from formic acid accumulation. Fomepizole is the preferred antidote to block alcohol dehydrogenase, with hemodialysis considered for severe cases.

#259 · Toxicology, Environmental Emergencies, and Anaphylaxis

What triad of findings suggests opioid toxicity, and what is the reversal agent?

Quick Reference

Miosis (pinpoint pupils), respiratory depression, and decreased level of consciousness form the classic opioid toxidrome. Naloxone is the reversal agent, though repeated doses or an infusion may be needed for long-acting opioids since naloxone's duration is shorter than many opioids.

#260 · Toxicology, Environmental Emergencies, and Anaphylaxis

A patient with tricyclic antidepressant overdose develops widened QRS on the electrocardiogram. What is the first-line treatment?

Quick Reference

Sodium bicarbonate is first-line, given as an intravenous bolus to both alkalinize serum and increase sodium load, which counteracts the sodium channel blockade responsible for QRS widening and arrhythmia risk. This should be given based on QRS widening, not waiting for arrhythmia to develop.

#261 · Toxicology, Environmental Emergencies, and Anaphylaxis

Why is activated charcoal not given routinely to every poisoning patient?

Quick Reference

Activated charcoal is most effective within roughly one to two hours of ingestion (longer for sustained-release formulations or drugs that slow gastric emptying), requires a protected airway or cooperative patient, and does not bind certain substances well (such as alcohols, metals like iron, and lithium). Its use should be individualized based on timing, substance, and airway safety rather than applied to every overdose.

#262 · Toxicology, Environmental Emergencies, and Anaphylaxis

What clinical picture suggests anticholinergic toxicity, and what mnemonic describes it?

Quick Reference

"Hot as a hare, blind as a bat, dry as a bone, red as a beet, and mad as a hatter" describes anticholinergic toxicity: hyperthermia, mydriasis, dry mucous membranes and skin, flushing, and delirium. This contrasts with cholinergic toxicity, which causes the opposite pattern of secretions and miosis.

#263 · Toxicology, Environmental Emergencies, and Anaphylaxis

A farm worker presents with excessive salivation, lacrimation, urination, defecation, and miosis after pesticide exposure. What is the toxidrome and antidote?

Quick Reference

This is a cholinergic toxidrome from organophosphate poisoning, treated with atropine to control secretions and pralidoxime to reactivate acetylcholinesterase if given early. Decontamination of clothing and skin is also essential to prevent ongoing exposure to staff and the patient.

#264 · Toxicology, Environmental Emergencies, and Anaphylaxis

Why is a normal initial acetaminophen level at a very early time point after ingestion potentially misleading?

Quick Reference

Levels drawn too early (before about 4 hours post-ingestion) may not reflect peak absorption, especially with extended-release formulations or co-ingestants that delay gastric emptying, so a single early level cannot exclude toxicity and should be repeated. This is a common trap of anchoring on one early low number.

#265 · Toxicology, Environmental Emergencies, and Anaphylaxis

What toxin should be suspected in a patient with a cherry-red skin appearance, headache, and altered mental status after exposure to a fire or faulty heating source?

Quick Reference

Carbon monoxide poisoning should be suspected, though the classic cherry-red skin finding is actually uncommon in practice, and diagnosis relies on carboxyhemoglobin level measurement with high-flow oxygen or hyperbaric oxygen considered for severe cases. Relying on the classic skin finding to make the diagnosis is a trap since it is often absent.

#266 · Toxicology, Environmental Emergencies, and Anaphylaxis

A patient ingests an unknown quantity of extended-release calcium channel blocker and develops bradycardia and hypotension. What treatment beyond calcium and atropine may be needed?

Quick Reference

High-dose insulin euglycemia therapy is a key treatment for severe calcium channel blocker toxicity, improving cardiac contractility by supporting myocardial metabolism, given alongside dextrose to maintain euglycemia. Relying on calcium and atropine alone is often insufficient in severe poisoning.

#267 · Toxicology, Environmental Emergencies, and Anaphylaxis

What is the significance of a normal anion gap versus an elevated anion gap in toxic ingestions, and what mnemonic lists elevated anion gap causes?

Quick Reference

MUDPILES (methanol, uremia, diabetic ketoacidosis, propylene glycol/paraldehyde, isoniazid/iron, lactic acidosis, ethylene glycol, salicylates) lists classic causes of elevated anion gap acidosis. Recognizing the pattern narrows the differential quickly in an undifferentiated poisoned or critically ill patient.

#268 · Toxicology, Environmental Emergencies, and Anaphylaxis

Why should a salicylate-poisoned patient not be intubated without careful consideration of ventilatory strategy?

Quick Reference

Salicylate-poisoned patients typically hyperventilate to compensate for a mixed respiratory alkalosis and metabolic acidosis, and if intubated, ventilator settings must maintain this same degree of hyperventilation, since normalizing their pCO₂ to a standard rate can cause rapid acidemia and clinical deterioration. Intubation should be avoided when possible and done cautiously if unavoidable.

#269 · Toxicology, Environmental Emergencies, and Anaphylaxis

A patient has QT prolongation and torsades de pointes after ingesting an unknown substance. What is the initial emergency treatment?

Quick Reference

Intravenous magnesium sulfate is the initial treatment for torsades de pointes regardless of the underlying cause, given rapidly in unstable patients, with correction of other electrolytes such as potassium as needed. Defibrillation is used if the patient becomes pulseless.

#270 · Toxicology, Environmental Emergencies, and Anaphylaxis

Why is co-ingestion an important consideration in every intentional overdose presentation?

Quick Reference

Patients presenting after intentional self-poisoning frequently take more than one substance, so a single identified drug or a reassuring level for one agent does not exclude a dangerous co-ingestant. Broad screening, collateral history, and a period of observation are important even when one substance seems clearly identified.

#271 · Toxicology, Environmental Emergencies, and Anaphylaxis

What is the mainstay treatment for iron overdose and why does activated charcoal not help?

Quick Reference

Deferoxamine chelation is used for significant iron toxicity, guided by serum iron levels and clinical severity, since iron is not adsorbed by activated charcoal at all. Whole bowel irrigation may be used instead to remove unabsorbed tablets from the gastrointestinal tract.

#272 · Toxicology, Environmental Emergencies, and Anaphylaxis

What are the core management priorities for severe hypothermia with an unstable cardiac rhythm?

Quick Reference

Gentle handling to avoid triggering ventricular fibrillation, active core rewarming, and continued resuscitation efforts are prioritized, since a severely hypothermic patient may appear clinically dead but can still be successfully resuscitated. The teaching that "a patient is not dead until warm and dead" reflects this principle.

#273 · Toxicology, Environmental Emergencies, and Anaphylaxis

A patient with heat stroke has a core temperature of 41 degrees Celsius and altered mental status. What distinguishes heat stroke from heat exhaustion, and what is the priority treatment?

Quick Reference

Heat stroke is defined by central nervous system dysfunction (confusion, seizure, coma) with severe hyperthermia, unlike heat exhaustion which has a normal mental status. Rapid active cooling, ideally with cold water immersion, is the priority and should not be delayed for other interventions.

#274 · Toxicology, Environmental Emergencies, and Anaphylaxis

What is the correct initial field management for a suspected pit viper snakebite, and what should be avoided?

Quick Reference

Immobilize the limb at heart level, keep the patient calm, and transport promptly to a facility with antivenom available. Incision and suction, tourniquets, and ice application should all be avoided since they do not reduce venom effects and can worsen tissue injury.

#275 · Toxicology, Environmental Emergencies, and Anaphylaxis

A diver develops joint pain, neurologic symptoms, and shortness of breath after a rapid ascent. What is the diagnosis and definitive treatment?

Quick Reference

This suggests decompression sickness from nitrogen bubble formation, and definitive treatment is recompression in a hyperbaric oxygen chamber. High-flow oxygen and keeping the patient supine during transport are important supportive measures while arranging transfer.

#276 · Toxicology, Environmental Emergencies, and Anaphylaxis

Why should frostbitten tissue not be rubbed or massaged, and what is the correct rewarming method?

Quick Reference

Rubbing frostbitten tissue causes further mechanical damage to already fragile, ice-crystal-laden cells. Rapid rewarming in warm water (approximately 37 to 39 degrees Celsius) is the correct approach, and only once there is no risk of refreezing during transport.

#277 · Toxicology, Environmental Emergencies, and Anaphylaxis

A patient rescued from cold water immersion appears to be in cardiac arrest. What special consideration applies to resuscitation and pronouncing death?

Quick Reference

Hypothermia can slow metabolic demand enough to allow prolonged cardiac arrest with intact neurologic recovery, so resuscitation should generally continue and active rewarming pursued before death is declared, unlike normothermic arrest. Prolonged downtime alone should not be used to withhold resuscitation in cold water drowning.

#278 · Toxicology, Environmental Emergencies, and Anaphylaxis

What historical and exam feature distinguishes a black widow spider bite from a brown recluse spider bite?

Quick Reference

Black widow bites cause painful muscle cramping and rigidity (neurotoxic effect) often with minimal local skin change, while brown recluse bites can cause a slowly enlarging necrotic skin lesion (cytotoxic effect) with less systemic muscular symptoms. Management differs since black widow envenomation may need antivenom or muscle relaxants while brown recluse care is mainly wound care.

#279 · Toxicology, Environmental Emergencies, and Anaphylaxis

What is the appropriate management approach for a jellyfish sting, and what common home remedy should be avoided?

Quick Reference

Rinsing with vinegar (for certain species) or hot water immersion can help inactivate remaining nematocysts and relieve pain, while fresh water rinsing or rubbing the area should be avoided since it can trigger further nematocyst discharge. The specific first aid can vary by jellyfish species and region.

#280 · Toxicology, Environmental Emergencies, and Anaphylaxis

A hiker at high altitude develops headache, nausea, and difficulty sleeping. What is the diagnosis and appropriate management?

Quick Reference

This describes acute mountain sickness, managed with halting further ascent, rest, hydration, and analgesics, with acetazolamide used for both prevention and treatment. Continuing to ascend despite symptoms risks progression to high altitude cerebral or pulmonary edema.

#281 · Toxicology, Environmental Emergencies, and Anaphylaxis

What clinical progression from acute mountain sickness signals a life-threatening emergency requiring immediate descent?

Quick Reference

The development of ataxia, severe confusion, or altered level of consciousness indicates high altitude cerebral edema, and shortness of breath at rest with crackles suggests high altitude pulmonary edema. Immediate descent is the most effective treatment for either, and should not be delayed for medication trials alone.

#282 · Toxicology, Environmental Emergencies, and Anaphylaxis

Why is passive external rewarming alone often insufficient for a patient with moderate to severe hypothermia?

Quick Reference

Passive rewarming (blankets, warm environment) relies on the patient's own heat production, which is inadequate once shivering thermogenesis fails at lower core temperatures, so active external and internal rewarming methods (warmed intravenous fluids, forced air warming, or more invasive techniques) are needed. Relying on blankets alone in severe hypothermia delays adequate rewarming.

#283 · Toxicology, Environmental Emergencies, and Anaphylaxis

A patient develops muscle cramps, heavy sweating, and normal mental status after exertion in the heat. What is the diagnosis and key treatment difference from heat stroke?

Quick Reference

This is heat exhaustion, distinguished from heat stroke by preserved mental status, and is treated with rest, oral or intravenous fluid and electrolyte replacement, and passive cooling rather than the aggressive rapid cooling required for heat stroke. Confusing the two changes the urgency and method of cooling.

#284 · Toxicology, Environmental Emergencies, and Anaphylaxis

What is the priority assessment in a near-drowning patient who appears well after rescue with no obvious respiratory distress?

Quick Reference

Even asymptomatic near-drowning patients should be observed for a period since delayed pulmonary edema or respiratory deterioration can occur hours later from aspirated fluid, so discharge immediately after a normal initial exam without any observation period is a trap. Oxygen saturation and respiratory status should be monitored over time before clearing the patient.

#285 · Toxicology, Environmental Emergencies, and Anaphylaxis

Why should alcohol-based rubs or friction methods not be used to rewarm frostbitten extremities?

Quick Reference

Alcohol and friction can cause additional chemical or mechanical injury to already damaged, fragile frostbitten tissue and do not provide effective controlled rewarming. Warm water immersion at a controlled temperature remains the correct method.

#286 · Toxicology, Environmental Emergencies, and Anaphylaxis

What is the single most important first-line medication in anaphylaxis, and what is the correct route and site?

Quick Reference

Intramuscular epinephrine into the anterolateral thigh is first-line and should be given immediately once anaphylaxis is recognized, without waiting to see if symptoms worsen. Delaying epinephrine in favor of antihistamines or steroids first is a critical and dangerous error.

#287 · Toxicology, Environmental Emergencies, and Anaphylaxis

What clinical criteria help identify anaphylaxis when a patient has only skin symptoms plus one other system involved?

Quick Reference

Anaphylaxis can be diagnosed with acute onset of skin or mucosal symptoms (hives, swelling, flushing) plus involvement of at least one other system such as respiratory compromise, hypotension, or persistent gastrointestinal symptoms, even without prior known allergen exposure. Waiting for full-blown shock or a known allergen history before treating is a common and dangerous trap.

#288 · Toxicology, Environmental Emergencies, and Anaphylaxis

Why should patients treated for anaphylaxis be observed for several hours after apparent resolution?

Quick Reference

Biphasic anaphylaxis, a recurrence of symptoms after initial improvement without further allergen exposure, can occur hours after the first reaction resolves, so observation periods (commonly several hours, longer for severe reactions) are recommended before discharge. Discharging immediately once symptoms resolve risks missing a biphasic reaction at home.

#289 · Toxicology, Environmental Emergencies, and Anaphylaxis

A patient on a beta-blocker has anaphylaxis that is not responding well to standard-dose epinephrine. What medication should be considered?

Quick Reference

Glucagon can be considered in beta-blocked patients with refractory anaphylaxis, since it can bypass the beta-receptor blockade to increase cardiac contractility and heart rate. Epinephrine should still be continued or increased as needed alongside glucagon rather than substituted out entirely.

#290 · Toxicology, Environmental Emergencies, and Anaphylaxis

What is the role of antihistamines and corticosteroids in acute anaphylaxis management?

Quick Reference

Antihistamines help relieve itching and hives, and corticosteroids may reduce the risk or severity of a biphasic reaction, but neither treats the airway, breathing, or circulatory compromise of anaphylaxis and neither should ever substitute for epinephrine. Treating with only an antihistamine in a patient with airway or hemodynamic involvement is inadequate and dangerous.

#291 · Toxicology, Environmental Emergencies, and Anaphylaxis

A patient given epinephrine for anaphylaxis remains hypotensive despite two appropriately spaced doses. What is the next step?

Quick Reference

Aggressive intravenous crystalloid boluses should be given for ongoing hypotension, and if refractory, an epinephrine infusion should be started rather than repeated intramuscular doses indefinitely. Refractory anaphylactic shock needs escalation to continuous intravenous vasopressor support.

#292 · Toxicology, Environmental Emergencies, and Anaphylaxis

Why might a patient with anaphylaxis develop stridor even before obvious skin findings appear?

Quick Reference

Laryngeal edema can develop early and progress rapidly, and airway symptoms do not always follow skin findings in a predictable order, so the airway must be continuously reassessed regardless of when other symptoms started. Waiting for hives to appear before considering airway compromise can delay recognition of a life-threatening presentation.

#293 · Toxicology, Environmental Emergencies, and Anaphylaxis

What discharge instruction is essential for every patient treated for anaphylaxis in the emergency department?

Quick Reference

Every patient should be prescribed or confirmed to have an epinephrine auto-injector, given education on how and when to use it, and referred for allergy follow-up to identify the trigger and consider further management. Discharging without ensuring access to an auto-injector leaves the patient unprotected for a recurrence.

#294 · Toxicology, Environmental Emergencies, and Anaphylaxis

Why is intramuscular epinephrine preferred over subcutaneous injection in anaphylaxis?

Quick Reference

Intramuscular injection into the thigh achieves faster and more reliable peak plasma epinephrine concentrations than subcutaneous injection, which can have delayed and erratic absorption, especially during shock when peripheral perfusion is reduced. This is why the older subcutaneous route teaching has been replaced.

#295 · Toxicology, Environmental Emergencies, and Anaphylaxis

A patient develops anaphylaxis from a food allergen but has mild symptoms that improve quickly after one dose of epinephrine. Is prolonged monitoring still necessary?

Quick Reference

Yes, even mild or rapidly improving reactions warrant an observation period because biphasic reactions and severity of the initial presentation do not perfectly predict recurrence, and food-triggered reactions have been associated with biphasic recurrence. Assuming a quick response means the risk has passed is a common misjudgment.

#296 · Toxicology, Environmental Emergencies, and Anaphylaxis

What positioning should be used for a hypotensive anaphylaxis patient, and what position should be avoided if the patient is vomiting?

Quick Reference

Supine positioning with legs elevated helps support venous return and blood pressure in a hypotensive patient, but if vomiting occurs, the patient should be placed in the recovery position to protect the airway rather than remaining flat on the back. Sitting a hypotensive patient fully upright can worsen venous return and precipitate cardiovascular collapse, a phenomenon linked to sudden death in some cases.

#297 · Toxicology, Environmental Emergencies, and Anaphylaxis

Why is repeat dosing of epinephrine every 5 to 15 minutes acceptable and expected rather than a sign of treatment failure?

Quick Reference

A single dose of intramuscular epinephrine may not fully control severe or ongoing anaphylaxis, and repeat dosing at 5 to 15 minute intervals is standard practice if symptoms persist, rather than escalating immediately to alternative agents without adequate epinephrine dosing first. Under-dosing epinephrine due to fear of side effects is a more common problem than over-dosing.

#298 · Toxicology, Environmental Emergencies, and Anaphylaxis

A patient with a history of idiopathic anaphylaxis asks why no trigger was ever identified. What should be explained about anaphylaxis without a clear cause?

Quick Reference

A meaningful proportion of anaphylaxis cases have no identifiable trigger despite thorough allergy workup, termed idiopathic anaphylaxis, and these patients still require an epinephrine auto-injector and an emergency action plan regardless of the uncertain cause. Lack of an identified trigger does not mean the risk of recurrence is lower.

#299 · Toxicology, Environmental Emergencies, and Anaphylaxis

Why should every anaphylaxis patient have a tryptase level considered, and when should it be drawn?

Quick Reference

Serum tryptase can support the diagnosis retrospectively, especially when the presentation is atypical or the cause is unclear, and is most useful when drawn within 1 to 2 hours of symptom onset since it rises quickly and falls back to baseline within hours. A normal tryptase does not exclude anaphylaxis, since levels can remain normal in food-triggered reactions even when the diagnosis is clinically clear.

#300 · Obstetric, Pediatric, and Vulnerable Populations

First trimester bleeding with hemodynamic instability: what is the immediate priority before any imaging?

Quick Reference

Resuscitate first: IV access, crystalloid, type and crossmatch, and call obstetrics or surgery immediately. A ruptured ectopic pregnancy can cause rapid hemorrhagic shock, so unstable patients go straight to the OR rather than waiting for a formal pelvic ultrasound.

#301 · Obstetric, Pediatric, and Vulnerable Populations

What beta-hCG discriminatory zone is used to expect a visible intrauterine pregnancy on transvaginal ultrasound?

Quick Reference

Around 1500 to 2000 IU/L on transvaginal ultrasound. Above this level with no intrauterine sac seen, ectopic pregnancy must be assumed until proven otherwise; below it, a normal early pregnancy can still be too small to see.

#302 · Obstetric, Pediatric, and Vulnerable Populations

Classic exam trap: a single low beta-hCG value rules out ectopic pregnancy.

Quick Reference

False. No single hCG value excludes ectopic pregnancy. Serial hCG trends (should roughly double every 48 to 72 hours in a normal early pregnancy) and ultrasound findings matter far more than one isolated number.

#303 · Obstetric, Pediatric, and Vulnerable Populations

Rh-negative woman with first trimester bleeding: what must you give regardless of the cause of bleeding?

Quick Reference

Rh immunoglobulin (WinRho) to prevent alloimmunization, even for a simple threatened miscarriage. The exact dose depends on gestational age, and this is a frequently missed step on the exam.

#304 · Obstetric, Pediatric, and Vulnerable Populations

Differentiate threatened, inevitable, incomplete, and complete miscarriage by cervical os and ultrasound findings.

Quick Reference

Threatened: closed os, viable pregnancy on ultrasound. Inevitable: open os, products not yet passed. Incomplete: open os, some products retained on ultrasound. Complete: closed os, empty uterus, with a history of passing tissue.

#305 · Obstetric, Pediatric, and Vulnerable Populations

Name the three classic risk factors for ectopic pregnancy that get tested repeatedly.

Quick Reference

Prior ectopic pregnancy, prior tubal surgery or pelvic inflammatory disease causing tubal scarring, and a current IUD in place. Assisted reproductive technology is also a recognized risk factor.

#306 · Obstetric, Pediatric, and Vulnerable Populations

Why is a transabdominal-only pelvic ultrasound often insufficient in an early pregnancy bleeding workup?

Quick Reference

Transabdominal ultrasound is less sensitive for a small early intrauterine sac and for a small ectopic mass. Transvaginal ultrasound is the correct first-line study when available for suspected ectopic pregnancy.

#307 · Obstetric, Pediatric, and Vulnerable Populations

What is a pseudosac on ultrasound and why does it matter?

Quick Reference

A pseudosac is a fluid collection in the endometrial cavity that can mimic an early intrauterine gestational sac in a patient who actually has an ectopic pregnancy. Look for a true double decidual sign or a yolk sac to confirm a real intrauterine pregnancy.

#308 · Obstetric, Pediatric, and Vulnerable Populations

Molar pregnancy: what clinical clues distinguish it from a normal early pregnancy with bleeding?

Quick Reference

Markedly elevated beta-hCG for gestational age, a uterus large for dates, hyperemesis, early pre-eclampsia signs before 20 weeks, and a classic snowstorm or cluster-of-grapes pattern on ultrasound with no fetus seen.

#309 · Obstetric, Pediatric, and Vulnerable Populations

What analgesic and anti-emetic considerations apply to a stable patient with threatened miscarriage being discharged?

Quick Reference

Acetaminophen for pain is preferred, since NSAIDs are typically avoided in pregnancy. Give clear return precautions for worsening bleeding, pain, dizziness, or fever, and arrange obstetric follow-up with repeat hCG or ultrasound.

#310 · Obstetric, Pediatric, and Vulnerable Populations

Ruptured corpus luteum cyst can mimic ectopic pregnancy: what history and exam clue helps differentiate them?

Quick Reference

Corpus luteum rupture usually occurs with a confirmed intrauterine pregnancy already seen on a prior ultrasound, and pain is often unilateral adnexal without significant vaginal bleeding, though hemoperitoneum can still cause instability.

#311 · Obstetric, Pediatric, and Vulnerable Populations

A patient with first trimester bleeding has a positive urine hCG but ultrasound shows no intrauterine or adnexal pregnancy and no free fluid. What is this called and what is the disposition?

Quick Reference

This is a pregnancy of unknown location. Disposition is close outpatient follow-up with serial hCG in 48 hours and repeat ultrasound if the patient remains stable, since ectopic pregnancy cannot yet be excluded.

#312 · Obstetric, Pediatric, and Vulnerable Populations

What physical exam finding raises concern for hemoperitoneum in a suspected ectopic pregnancy?

Quick Reference

Cervical motion tenderness, adnexal tenderness or a mass, and peritoneal signs such as rebound or referred shoulder tip pain from diaphragmatic irritation by blood. Shoulder tip pain in a pregnant patient with abdominal pain is a classic tested clue.

#313 · Obstetric, Pediatric, and Vulnerable Populations

What is the medical management option for a confirmed small unruptured ectopic pregnancy in a stable patient?

Quick Reference

Methotrexate, provided the patient meets criteria: hemodynamically stable, no fetal cardiac activity, hCG below the institution's threshold, ectopic mass usually under 3.5 cm, and reliable follow-up for serial hCG monitoring.

#314 · Obstetric, Pediatric, and Vulnerable Populations

Define the diagnostic criteria for pre-eclampsia.

Quick Reference

New-onset hypertension (systolic 140 or higher, or diastolic 90 or higher, on two occasions) after 20 weeks gestation, plus either proteinuria or evidence of end-organ dysfunction such as renal, liver, neurologic, hematologic, or uteroplacental involvement.

#315 · Obstetric, Pediatric, and Vulnerable Populations

What defines severe features of pre-eclampsia?

Quick Reference

Systolic blood pressure of 160 or higher or diastolic 110 or higher, thrombocytopenia, impaired liver function or severe right upper quadrant pain, renal insufficiency, pulmonary edema, or new visual or cerebral symptoms such as a headache unresponsive to analgesia.

#316 · Obstetric, Pediatric, and Vulnerable Populations

Trap: proteinuria is required to diagnose pre-eclampsia.

Quick Reference

False. Current criteria allow diagnosis without proteinuria if other end-organ dysfunction is present, since some women progress to severe disease without ever spilling significant protein in the urine.

#317 · Obstetric, Pediatric, and Vulnerable Populations

What is the first-line antihypertensive approach for acute severe hypertension in pre-eclampsia in the ED?

Quick Reference

IV labetalol or IV hydralazine, or oral nifedipine, are first-line agents for acute severe hypertension of 160/110 or higher, used to reduce stroke risk while awaiting obstetric transfer.

#318 · Obstetric, Pediatric, and Vulnerable Populations

What is the drug of choice for eclamptic seizure prevention and treatment?

Quick Reference

Magnesium sulfate, not phenytoin or benzodiazepines, is the drug of choice both to treat an active eclamptic seizure and to prevent seizures in severe pre-eclampsia.

#319 · Obstetric, Pediatric, and Vulnerable Populations

What is an early sign of magnesium toxicity to monitor for during infusion?

Quick Reference

Loss of deep tendon reflexes is the earliest sign, followed by respiratory depression and then cardiac arrest at higher levels. Monitor reflexes, respiratory rate, and urine output throughout the infusion.

#320 · Obstetric, Pediatric, and Vulnerable Populations

What is the antidote for magnesium toxicity?

Quick Reference

IV calcium gluconate is the antidote for magnesium sulfate toxicity, given for absent deep tendon reflexes or respiratory depression.

#321 · Obstetric, Pediatric, and Vulnerable Populations

What are the components of HELLP syndrome?

Quick Reference

Hemolysis, Elevated Liver enzymes, and Low Platelets. It is a severe variant of pre-eclampsia and can occur even with normal blood pressure, which is another common exam trap.

#322 · Obstetric, Pediatric, and Vulnerable Populations

What is the only definitive treatment for pre-eclampsia?

Quick Reference

Delivery of the placenta. Medical management controls blood pressure and prevents seizures but does not cure the underlying disease; timing of delivery depends on gestational age and disease severity.

#323 · Obstetric, Pediatric, and Vulnerable Populations

Trap: pre-eclampsia only occurs during pregnancy, never after delivery.

Quick Reference

False. Eclampsia and pre-eclampsia can present up to 4 to 6 weeks postpartum, so a new severe headache or hypertension in a recent postpartum patient must still trigger the same full workup.

#324 · Obstetric, Pediatric, and Vulnerable Populations

What chief complaint should raise suspicion for pre-eclampsia even without known hypertension?

Quick Reference

Right upper quadrant or epigastric pain in a pregnant patient, especially after 20 weeks, can represent hepatic capsule distension from HELLP syndrome and should prompt a blood pressure check, urinalysis, and liver enzymes.

#325 · Obstetric, Pediatric, and Vulnerable Populations

Why should NSAIDs be avoided in a patient with confirmed pre-eclampsia?

Quick Reference

NSAIDs can worsen hypertension and impair renal function, both of which are already compromised in pre-eclampsia, so acetaminophen is preferred for analgesia in this population.

#326 · Obstetric, Pediatric, and Vulnerable Populations

What fetal risk should be explained when counseling a patient with pre-eclampsia?

Quick Reference

Uteroplacental insufficiency can cause intrauterine growth restriction, oligohydramnios, placental abruption, and fetal distress, which is why fetal monitoring accompanies maternal stabilization.

#327 · Obstetric, Pediatric, and Vulnerable Populations

Distinguish chronic hypertension, gestational hypertension, and pre-eclampsia by timing.

Quick Reference

Chronic hypertension predates pregnancy or is diagnosed before 20 weeks. Gestational hypertension arises after 20 weeks without proteinuria or end-organ dysfunction. Pre-eclampsia is hypertension after 20 weeks with proteinuria or end-organ involvement.

#328 · Obstetric, Pediatric, and Vulnerable Populations

What age cutoff is most critical for triggering an aggressive fever workup in infants?

Quick Reference

Infants under 90 days, and especially under 28 days, with fever require a full septic workup because clinical exam is unreliable for detecting serious bacterial infection at that age.

#329 · Obstetric, Pediatric, and Vulnerable Populations

What classic bacteria must be covered in empiric antibiotics for neonatal fever?

Quick Reference

Group B Streptococcus, E. coli, and Listeria monocytogenes are the classic neonatal pathogens, guiding empiric ampicillin plus a third-generation cephalosporin or an aminoglycoside.

#330 · Obstetric, Pediatric, and Vulnerable Populations

What is the purpose of the Step-by-Step or Rochester criteria?

Quick Reference

These are validated low-risk criteria used to identify febrile infants 29 to 90 days old who may be safely managed without lumbar puncture or admission, based on a normal exam, normal labs, and no focal source.

#331 · Obstetric, Pediatric, and Vulnerable Populations

Trap: a fully immunized, well-appearing 18-month-old with fever and no source needs a full septic workup.

Quick Reference

False. In the post-conjugate-vaccine era, well-appearing, fully immunized febrile toddlers with no focal source generally do not need blind blood cultures or a septic workup; clinical judgment and a thorough exam guide management instead.

#332 · Obstetric, Pediatric, and Vulnerable Populations

Define fever without a source and describe the key age-based branch point in management.

Quick Reference

Fever without an identifiable source after history and exam; management differs sharply by age, with infants under 90 days needing an aggressive workup while older well-appearing immunized children can often be safely observed.

#333 · Obstetric, Pediatric, and Vulnerable Populations

Why is urine the most commonly missed source of fever in young children?

Quick Reference

Urinary tract infection can present with fever alone and no localizing signs, especially in girls under 2 and uncircumcised boys under 1, so a urine sample should be obtained in this fever-without-source population.

#334 · Obstetric, Pediatric, and Vulnerable Populations

How reliable are Kernig's and Brudzinski's signs in infants?

Quick Reference

These meningismus signs are unreliable in infants under 12 months, since the neurologic exam for meningitis is far less sensitive at that age; a bulging fontanelle or unexplained irritability is more useful.

#335 · Obstetric, Pediatric, and Vulnerable Populations

Fever plus petechiae in a child: what must be ruled out urgently?

Quick Reference

Invasive meningococemia. Petechiae below the nipple line, an ill appearance, or rapidly spreading petechiae warrant urgent blood cultures and empiric antibiotics without delay.

#336 · Obstetric, Pediatric, and Vulnerable Populations

What triggers the diagnostic criteria for Kawasaki disease?

Quick Reference

Fever for 5 or more days plus 4 of 5 features: conjunctivitis, oral mucosal changes, a polymorphous rash, extremity changes, and cervical lymphadenopathy. A missed diagnosis risks coronary artery aneurysm.

#337 · Obstetric, Pediatric, and Vulnerable Populations

Why does antipyretic dosing matter on the exam?

Quick Reference

Dosing is weight-based (acetaminophen 10 to 15 mg/kg, ibuprofen 5 to 10 mg/kg), and antipyretics are given mainly for comfort, since fever height itself does not reliably predict serious bacterial illness.

#338 · Obstetric, Pediatric, and Vulnerable Populations

Trap: the height of a fever alone predicts serious bacterial infection.

Quick Reference

False. Degree of fever correlates poorly with likelihood of serious bacterial infection; clinical appearance, age, and specific risk factors matter far more than the exact temperature reading.

#339 · Obstetric, Pediatric, and Vulnerable Populations

How should a simple febrile seizure be classified and what is the disposition?

Quick Reference

Simple febrile seizures are generalized, last under 15 minutes, and do not recur in 24 hours. In a well child over 6 months, they generally do not require neuroimaging or lumbar puncture and can be discharged once the fever source is addressed; complex features warrant further workup.

#340 · Obstetric, Pediatric, and Vulnerable Populations

What is the occult bacteremia workup consideration in the 3 to 36 month age group?

Quick Reference

In a well-appearing, immunized child this age, occult bacteremia risk is low, and blood cultures are not routinely needed, but a source should still be sought by checking ears, throat, urine, and skin.

#341 · Obstetric, Pediatric, and Vulnerable Populations

Why is lumbar puncture often performed before antibiotics in suspected neonatal sepsis, when feasible?

Quick Reference

Antibiotics can sterilize CSF quickly and obscure culture results, so if LP can be performed promptly without meaningfully delaying antibiotics, it should be done first to preserve diagnostic yield.

#342 · Obstetric, Pediatric, and Vulnerable Populations

What classic fracture pattern raises suspicion for non-accidental trauma in infants?

Quick Reference

Metaphyseal corner or bucket-handle fractures and posterior rib fractures are highly specific for non-accidental trauma in infants, since these are unusual accidental injury patterns at that age.

#343 · Obstetric, Pediatric, and Vulnerable Populations

Trap: bruising location is not useful for distinguishing abuse from accidental injury.

Quick Reference

False. Bruising over soft, padded, or normally protected areas such as the ears, neck, buttocks, genitals, or torso in a pre-mobile infant is far more suspicious than bruising over bony prominences like shins or the forehead, which is typical of accidental play injury.

#344 · Obstetric, Pediatric, and Vulnerable Populations

What is the purpose of the TEN-4-FACESp bruising rule?

Quick Reference

A validated clinical decision rule that flags bruising on the Torso, Ears, or Neck in a child under 4 years, or bruising anywhere on an infant under 4 to 5 months, as high risk for abuse and warranting further workup.

#345 · Obstetric, Pediatric, and Vulnerable Populations

What is a physician's mandatory reporting obligation for suspected child abuse?

Quick Reference

Physicians have a legal duty to report suspected child abuse or neglect to child protection services, and this duty exists on reasonable suspicion, not proof, overriding the usual rules of confidentiality.

#346 · Obstetric, Pediatric, and Vulnerable Populations

What documentation standard applies to suspected abuse encounters?

Quick Reference

Use direct quotes from the patient and caregiver, describe injuries objectively with measurements and photographs, and avoid conclusory language, since the chart may later become a legal document.

#347 · Obstetric, Pediatric, and Vulnerable Populations

Explain the sentinel injury concept in infants.

Quick Reference

A sentinel injury is a minor but concerning injury, such as a facial bruise or intraoral injury, in a pre-mobile infant that, if missed, often precedes a subsequent more severe abusive injury; it should always trigger a full skeletal survey.

#348 · Obstetric, Pediatric, and Vulnerable Populations

What is the indication for a skeletal survey and appropriate follow-up in suspected abuse?

Quick Reference

A full skeletal survey is indicated in any child under 2 with suspected physical abuse, since occult fractures at different healing stages are common. A repeat survey in about 2 weeks increases sensitivity for subtle or healing fractures.

#349 · Obstetric, Pediatric, and Vulnerable Populations

What is the classic triad of shaken baby syndrome, also called abusive head trauma?

Quick Reference

Subdural hemorrhage, retinal hemorrhages, and encephalopathy, often without any external signs of trauma, in an infant with a vague or inconsistent history.

#350 · Obstetric, Pediatric, and Vulnerable Populations

How should intimate partner violence screening be approached in the ED?

Quick Reference

Screen in a private setting without the partner present, using direct nonjudgmental questions. Routine universal screening is recommended for women presenting with injury patterns or vague, unexplained symptoms.

#351 · Obstetric, Pediatric, and Vulnerable Populations

What lethality risk factors should be asked about in suspected intimate partner violence?

Quick Reference

Escalating frequency or severity of violence, a history of strangulation, firearm access in the home, and threats to kill are the strongest predictors of homicide risk and should prompt safety planning.

#352 · Obstetric, Pediatric, and Vulnerable Populations

What is the exam trap regarding strangulation injuries?

Quick Reference

Absence of visible neck marks does not rule out dangerous strangulation. Internal injury such as laryngeal, vascular, or brain hypoxic injury can occur without external bruising, so any strangulation history warrants a thorough exam and a low threshold for imaging.

#353 · Obstetric, Pediatric, and Vulnerable Populations

What are red flags for elder abuse on physical exam?

Quick Reference

Unexplained weight loss, poor hygiene, pressure ulcers, bruising in various stages of healing, and a caregiver who answers all questions for the patient or is reluctant to leave the room.

#354 · Obstetric, Pediatric, and Vulnerable Populations

What is the safety and confidentiality principle for a disclosed case of intimate partner violence?

Quick Reference

Never confront or interview the patient in front of the suspected abuser. Offer resources such as a shelter, hotline, or safety plan without forcing action, and document findings carefully, since the patient may not be ready to leave the relationship immediately.

#355 · Obstetric, Pediatric, and Vulnerable Populations

Trap: a child's explanation that matches the visible injury excludes abuse.

Quick Reference

False. A plausible-sounding explanation does not exclude abuse if the child's developmental capability does not match the stated mechanism, such as a non-mobile infant said to have rolled off a bed. Developmental stage must always be checked against the story given.

#356 · Shock, Sepsis, and Thromboembolic Disease

Name the four classes of shock.

Quick Reference

Hypovolemic, cardiogenic, obstructive, and distributive shock, with distributive shock including septic, anaphylactic, and neurogenic causes. Each class has a distinct hemodynamic pattern that guides treatment.

#357 · Shock, Sepsis, and Thromboembolic Disease

What hemodynamic pattern distinguishes cardiogenic from hypovolemic shock?

Quick Reference

Cardiogenic shock shows high filling pressures with low cardiac output, described as wet and cold, while hypovolemic shock shows low filling pressures with low cardiac output, described as dry and cold. This distinction changes fluid versus inotrope decisions.

#358 · Shock, Sepsis, and Thromboembolic Disease

What is the classic hemodynamic pattern of early septic distributive shock?

Quick Reference

Warm shock, with low systemic vascular resistance and often normal or high cardiac output, warm extremities, and a wide pulse pressure, before later progressing to a cold, hypodynamic phase as the patient decompensates.

#359 · Shock, Sepsis, and Thromboembolic Disease

Trap: tachycardia is always present early in hemorrhagic shock.

Quick Reference

False, especially in young fit patients or those on beta-blockers. Compensatory mechanisms can mask tachycardia until decompensation is well underway, so blood pressure and heart rate alone should not be relied on to exclude significant blood loss.

#360 · Shock, Sepsis, and Thromboembolic Disease

What is the estimated blood loss and vital sign pattern for Class III hemorrhagic shock?

Quick Reference

Roughly 30 to 40 percent blood volume loss, with hypotension, marked tachycardia, and altered mental status. This is the point at which compensatory mechanisms begin to fail and blood pressure finally drops.

#361 · Shock, Sepsis, and Thromboembolic Disease

What is the initial fluid resuscitation choice in undifferentiated shock?

Quick Reference

Isotonic crystalloid, such as normal saline or a balanced solution like Ringer's lactate, given as a bolus with reassessment after each bolus. The underlying cause then dictates further therapy such as blood products, vasopressors, or source control.

#362 · Shock, Sepsis, and Thromboembolic Disease

What is the most reliable single clinical predictor of significant dehydration in a child?

Quick Reference

No single sign is best alone; capillary refill time and overall clinical impression, combined into scores like the WHO or Clinical Dehydration Scale, are more reliable than any one vital sign. Prolonged capillary refill, poor skin turgor, and abnormal breathing pattern together best predict significant dehydration.

#363 · Shock, Sepsis, and Thromboembolic Disease

When is oral rehydration therapy indicated in a dehydrated child?

Quick Reference

Mild to moderate dehydration in a child who can tolerate oral intake should be managed with oral rehydration solution first, reserving IV fluids for severe dehydration, shock, or failure of oral intake.

#364 · Shock, Sepsis, and Thromboembolic Disease

What is the role of lactate in shock assessment?

Quick Reference

Elevated lactate reflects tissue hypoperfusion and anaerobic metabolism. Trending lactate clearance over hours is used as a marker of resuscitation adequacy, not just as a single diagnostic cutoff at one point in time.

#365 · Shock, Sepsis, and Thromboembolic Disease

What is the hemodynamic signature and cause of neurogenic shock?

Quick Reference

Hypotension with bradycardia, not tachycardia, due to loss of sympathetic tone below a spinal cord injury. Treat with fluids first, then vasopressors with alpha-agonist activity if needed, distinguishing it clearly from hypovolemic shock.

#366 · Shock, Sepsis, and Thromboembolic Disease

What causes should be considered for obstructive shock?

Quick Reference

Tension pneumothorax, cardiac tamponade, and massive pulmonary embolism. Each impairs cardiac filling or output mechanically and requires a specific procedural fix rather than fluids alone.

#367 · Shock, Sepsis, and Thromboembolic Disease

What is the current practical application of early goal-directed therapy in septic shock?

Quick Reference

Early recognition, prompt broad-spectrum antibiotics within the first hour, and adequate fluid resuscitation guided by careful clinical reassessment remain the pillars, since strict early goal-directed therapy protocols were not shown to outperform usual careful care in later trials.

#368 · Shock, Sepsis, and Thromboembolic Disease

What is the first-line vasopressor in septic shock?

Quick Reference

Norepinephrine is first-line for septic shock, titrated to a mean arterial pressure target commonly around 65 mmHg, typically started after or alongside adequate fluid resuscitation.

#369 · Shock, Sepsis, and Thromboembolic Disease

Trap: a normal blood pressure excludes shock.

Quick Reference

False. Blood pressure is a late finding in compensated shock. Look instead for tachycardia, narrowed pulse pressure, delayed capillary refill, altered mentation, and rising lactate, which can all signal shock despite a normal blood pressure.

#370 · Shock, Sepsis, and Thromboembolic Disease

What fluid choice caution applies during large-volume resuscitation?

Quick Reference

Excessive normal saline can cause hyperchloremic metabolic acidosis. Balanced crystalloids are increasingly preferred for large-volume resuscitation to avoid this iatrogenic acid-base disturbance.

#371 · Shock, Sepsis, and Thromboembolic Disease

What is the empiric antibiotic choice for suspected bacterial meningitis in an adult?

Quick Reference

Vancomycin plus a third-generation cephalosporin, such as ceftriaxone or cefotaxime, covers the major pathogens. Add ampicillin if there is a *Listeria* risk, such as age over 50 or immunocompromise, and give dexamethasone before or with the first antibiotic dose.

#372 · Shock, Sepsis, and Thromboembolic Disease

Why give dexamethasone before or with the first dose of antibiotics in suspected bacterial meningitis?

Quick Reference

Steroids given after antibiotics lose their benefit of reducing neurologic complications and mortality, particularly in pneumococcal meningitis, so the timing relative to the antibiotic dose genuinely matters.

#373 · Shock, Sepsis, and Thromboembolic Disease

What clinical clue for necrotizing fasciitis outweighs a normal-appearing overlying skin exam?

Quick Reference

Pain out of proportion to visible skin findings, along with fever and systemic toxicity, should raise suspicion even when the overlying skin looks unremarkable. This mismatch is a classic tested trap.

#374 · Shock, Sepsis, and Thromboembolic Disease

What is the purpose and limitation of the LRINEC score?

Quick Reference

A scoring system using lab values such as CRP, WBC, hemoglobin, sodium, creatinine, and glucose to help identify necrotizing fasciitis, but it has imperfect sensitivity and should never replace surgical consultation when clinical suspicion is high.

#375 · Shock, Sepsis, and Thromboembolic Disease

What distinguishes management of cellulitis from an abscess?

Quick Reference

Cellulitis without a fluctuant collection is treated with antibiotics alone, while an abscess requires incision and drainage as primary therapy, with antibiotics as an adjunct especially in high-risk patients or larger lesions.

#376 · Shock, Sepsis, and Thromboembolic Disease

What severity tool is commonly used for community-acquired pneumonia in the ED?

Quick Reference

CURB-65, which stands for Confusion, Urea, Respiratory rate, Blood pressure, and age 65 or over, helps guide the decision between outpatient treatment, admission, or ICU-level care.

#377 · Shock, Sepsis, and Thromboembolic Disease

Trap: a normal chest x-ray excludes pneumonia.

Quick Reference

False. Early pneumonia, dehydration, or neutropenia can produce a falsely normal or minimally abnormal chest x-ray, so clinical findings should still guide treatment when suspicion is high.

#378 · Shock, Sepsis, and Thromboembolic Disease

What empiric antibiotic consideration applies to a septic patient with an unknown source and a possible line infection?

Quick Reference

Broad-spectrum antibiotics should cover gram-positive and gram-negative organisms and consider MRSA coverage with vancomycin if a central line or known MRSA risk factor is present, alongside prompt source control efforts such as line removal.

#379 · Shock, Sepsis, and Thromboembolic Disease

What classic triad should raise concern for tick-borne illness in the correct season and region?

Quick Reference

Fever, a rash such as the classic erythema migrans of Lyme disease, and joint or neurologic symptoms after a known tick exposure. Regional risk and season, generally spring through fall, should shape the differential.

#380 · Shock, Sepsis, and Thromboembolic Disease

What is the treatment window for oseltamivir in influenza and who benefits most?

Quick Reference

It is most effective when started within 48 hours of symptom onset, with the greatest benefit in high-risk patients such as the elderly, pregnant patients, immunocompromised patients, and those with chronic disease, though it can still be considered later in severely ill or hospitalized patients.

#381 · Shock, Sepsis, and Thromboembolic Disease

What clinical clue combination should raise suspicion for endocarditis in a febrile patient?

Quick Reference

A new murmur, fever, and a predisposing risk factor such as IV drug use, a prosthetic valve, or recent dental work should prompt blood cultures before antibiotics and consideration of echocardiography.

#382 · Shock, Sepsis, and Thromboembolic Disease

What triggers toxic shock syndrome and what is its classic presentation?

Quick Reference

A retained foreign body such as a tampon, nasal packing, or wound packing with *Staphylococcus aureus* or *Streptococcus* toxin production causes fever, a diffuse macular rash, hypotension, and multi-organ involvement. Removal of the source is essential alongside antibiotics.

#383 · Shock, Sepsis, and Thromboembolic Disease

Trap: antibiotics alone are sufficient source control for an abscess or infected retained foreign body.

Quick Reference

False. Drainage or removal of the source, such as abscess drainage, foreign body removal, or debridement of necrotic tissue, is required, since antibiotics without source control frequently fail.

#384 · Shock, Sepsis, and Thromboembolic Disease

What is the current consensus definition of sepsis?

Quick Reference

Life-threatening organ dysfunction caused by a dysregulated host response to infection, generally identified by an acute rise in SOFA score of 2 or more points, a shift away from the older SIRS-only definition.

#385 · Shock, Sepsis, and Thromboembolic Disease

What is the purpose of the Wells score for pulmonary embolism?

Quick Reference

A validated clinical prediction rule that stratifies pretest probability of pulmonary embolism into categories such as low, moderate, high, or PE likely versus unlikely, guiding whether D-dimer testing or direct imaging is the appropriate next step.

#386 · Shock, Sepsis, and Thromboembolic Disease

What is the purpose and correct use of the PERC rule?

Quick Reference

The Pulmonary Embolism Rule-out Criteria are used only in patients already judged low pretest probability. If all 8 criteria are negative, PE can be excluded without any further testing, including D-dimer.

#387 · Shock, Sepsis, and Thromboembolic Disease

Trap: the PERC rule can be applied to any patient presenting with chest pain.

Quick Reference

False. PERC only applies after the clinician has already determined the patient is low pretest probability. Applying it to moderate or high-probability patients is a common and dangerous exam trap.

#388 · Shock, Sepsis, and Thromboembolic Disease

What test characteristic determines the clinical usefulness of D-dimer?

Quick Reference

D-dimer has high sensitivity but low specificity, making it useful to rule out PE or DVT in low-to-moderate probability patients when negative, but a positive result is nonspecific and requires imaging for confirmation.

#389 · Shock, Sepsis, and Thromboembolic Disease

What is the rationale for an age-adjusted D-dimer cutoff?

Quick Reference

D-dimer rises naturally with age, causing more false positives in older patients. Using age times 10 nanograms per mL, for patients over 50, as the cutoff instead of a fixed value improves specificity without losing sensitivity.

#390 · Shock, Sepsis, and Thromboembolic Disease

What is the first-line imaging test for suspected DVT?

Quick Reference

Compression ultrasound of the lower extremity, which is highly sensitive and specific for proximal DVT and avoids radiation exposure.

#391 · Shock, Sepsis, and Thromboembolic Disease

What is the role and limitation of CT pulmonary angiography?

Quick Reference

It is the primary confirmatory imaging test for PE with high sensitivity and specificity, but it requires IV contrast and radiation, so it should be reserved for patients with at least moderate pretest probability or a positive D-dimer.

#392 · Shock, Sepsis, and Thromboembolic Disease

How is massive, or high-risk, PE defined and what immediate treatment should be considered?

Quick Reference

PE causing sustained hypotension or shock. These patients may need systemic thrombolysis if no major contraindication exists, since mortality is high and rapid reduction of clot burden can be lifesaving.

#393 · Shock, Sepsis, and Thromboembolic Disease

How is submassive PE defined?

Quick Reference

PE with right ventricular strain or dysfunction, shown by elevated troponin or BNP or RV dilation on echo or CT, but without frank hypotension. Management is individualized and thrombolysis is more selectively considered.

#394 · Shock, Sepsis, and Thromboembolic Disease

Trap: a normal blood pressure excludes a clinically significant PE.

Quick Reference

False. Right ventricular strain and hypoxia can be present with a normal systemic blood pressure in submassive PE, so vital signs alone should not be used to exclude a hemodynamically significant clot.

#395 · Shock, Sepsis, and Thromboembolic Disease

What is the first-line anticoagulation choice for a stable, newly diagnosed DVT or PE without major renal or malignancy considerations?

Quick Reference

A direct oral anticoagulant such as apixaban or rivaroxaban is now generally first-line for outpatient treatment of stable VTE, replacing routine bridging with heparin and warfarin in most patients.

#396 · Shock, Sepsis, and Thromboembolic Disease

What criteria support outpatient management of an uncomplicated PE?

Quick Reference

Hemodynamic stability, low bleeding risk, good social support, reliable follow-up, and a low-risk score such as the PESI or simplified PESI support safe outpatient treatment rather than automatic admission.

#397 · Shock, Sepsis, and Thromboembolic Disease

What is the trap regarding Homans' sign in suspected DVT?

Quick Reference

Homans' sign, calf pain on dorsiflexion, is neither sensitive nor specific for DVT and should not be relied upon to rule the diagnosis in or out. It is largely historical and not evidence-based.

#398 · Shock, Sepsis, and Thromboembolic Disease

What high-risk Wells DVT feature should prompt urgent imaging even while a D-dimer result is pending?

Quick Reference

Active cancer, recent immobilization or surgery, and a swollen calf more than 3 cm larger than the other leg all raise pretest probability enough that imaging should proceed even when D-dimer results are being awaited or are borderline.

#399 · Airway Management, Procedures, and Ophthalmology

What is the purpose of the LEMON mnemonic?

Quick Reference

A structured difficult airway prediction tool: Look externally, Evaluate the 3-3-2 rule, Mallampati score, Obstruction, and Neck mobility, used before attempting intubation to anticipate difficulty.

#400 · Airway Management, Procedures, and Ophthalmology

What is the core principle of rapid sequence intubation?

Quick Reference

Simultaneous administration of an induction agent and a paralytic without bag-mask ventilation in between, to secure the airway quickly while minimizing aspiration risk in a patient assumed to have a full stomach.

#401 · Airway Management, Procedures, and Ophthalmology

What contraindications to succinylcholine should be known cold?

Quick Reference

Hyperkalemia risk conditions including burns after 24 hours, crush injury, prolonged immobility, denervation injuries, and neuromuscular disease, plus a personal or family history of malignant hyperthermia.

#402 · Airway Management, Procedures, and Ophthalmology

What is the key practical difference between rocuronium and succinylcholine?

Quick Reference

Rocuronium has a longer duration of paralysis than succinylcholine but avoids the hyperkalemia risk, and its effect can now be reversed rapidly with sugammadex if available.

#403 · Airway Management, Procedures, and Ophthalmology

What hemodynamic advantage does ketamine offer for induction in RSI?

Quick Reference

Ketamine generally maintains or increases blood pressure and heart rate through catecholamine release, making it a preferred induction agent in hypotensive or shock patients compared to propofol or etomidate.

#404 · Airway Management, Procedures, and Ophthalmology

What is the concern with etomidate in septic patients?

Quick Reference

Etomidate can cause transient adrenal suppression, a theoretical concern in septic shock, though many guidelines still consider a single induction dose acceptable given its overall hemodynamic stability.

#405 · Airway Management, Procedures, and Ophthalmology

What is the current evidence status for cricoid pressure, or the Sellick maneuver?

Quick Reference

Evidence for cricoid pressure preventing aspiration is weak, and it can worsen the laryngoscopic view. Many current practices no longer apply it routinely, or release it if it impairs the view.

#406 · Airway Management, Procedures, and Ophthalmology

What predicts difficult bag-mask ventilation?

Quick Reference

Beard, obesity, older age, toothlessness, a history of snoring or sleep apnea, and stiff lungs with poor compliance all predict difficult mask ventilation, which matters because a failed intubation combined with unmanageable BMV is the most dangerous scenario.

#407 · Airway Management, Procedures, and Ophthalmology

What is the next step in the failed airway algorithm after failed intubation attempts when oxygenation is still adequate?

Quick Reference

Continue attempts with an alternate device or technique, such as video laryngoscopy, a bougie, or a supraglottic airway, while maintaining oxygenation between attempts, following a pre-planned difficult airway algorithm rather than repeating the same failed technique.

#408 · Airway Management, Procedures, and Ophthalmology

What is the definitive next step in a can't-intubate, can't-oxygenate scenario?

Quick Reference

Emergency surgical airway, or cricothyrotomy, must be performed without delay. This is a true crash airway scenario where continued attempts at intubation waste critical time.

#409 · Airway Management, Procedures, and Ophthalmology

What is the role of waveform capnography after intubation?

Quick Reference

Continuous waveform capnography is the gold standard confirmation of correct endotracheal tube placement and ongoing ventilation, more reliable than auscultation or chest rise alone, and it also detects accidental tube dislodgement immediately.

#410 · Airway Management, Procedures, and Ophthalmology

What is the goal and technique for pre-oxygenation before RSI?

Quick Reference

Achieve as close to 100 percent denitrogenation as possible using high-flow oxygen or a non-rebreather mask for at least 3 minutes, with consideration of apneic oxygenation via nasal cannula during the apneic period to extend safe apnea time.

#411 · Airway Management, Procedures, and Ophthalmology

Trap: a Mallampati score alone reliably predicts a difficult airway.

Quick Reference

False. The Mallampati score alone has limited predictive value and should be combined with other features such as mouth opening, thyromental distance, neck mobility, and obstruction, as in the full LEMON approach, for a more accurate assessment.

#412 · Airway Management, Procedures, and Ophthalmology

What is the indication for awake intubation?

Quick Reference

Reserved for a predicted very difficult airway where paralysis could lead to a can't-intubate, can't-ventilate scenario. Topical anesthesia and light sedation allow the patient to maintain spontaneous ventilation while the airway is secured, often using a fiberoptic approach.

#413 · Airway Management, Procedures, and Ophthalmology

What is the principle behind post-intubation sedation and analgesia?

Quick Reference

Provide both analgesia and sedation, not paralytic alone, after intubation, since paralytics without adequate sedation leave a conscious, paralyzed, and terrified patient. This is a common oversight after a busy RSI.

#414 · Airway Management, Procedures, and Ophthalmology

What is the core monitoring requirement for procedural sedation?

Quick Reference

Continuous pulse oximetry, capnography where available, blood pressure monitoring, and dedicated personnel monitoring the patient separately from the proceduralist throughout the sedation and recovery period.

#415 · Airway Management, Procedures, and Ophthalmology

What are the dosing route options for ketamine dissociative sedation?

Quick Reference

IV, typically around 1 to 1.5 mg/kg, or IM, typically around 4 mg/kg, can be used. IM is useful when IV access is difficult, particularly in pediatric procedural sedation.

#416 · Airway Management, Procedures, and Ophthalmology

What is a classic ketamine adverse effect to counsel families about, and how is it managed?

Quick Reference

An emergence reaction, with agitation, hallucinations, or vivid dreams, can occur during recovery. Keeping the recovery environment calm and quiet reduces its incidence, and benzodiazepines can treat a severe reaction if needed.

#417 · Airway Management, Procedures, and Ophthalmology

What is the fasting guideline nuance for emergency procedural sedation?

Quick Reference

Unlike elective anesthesia, strict fasting times are not an absolute requirement in the emergency setting. Aspiration risk is weighed against the urgency of the procedure, with recent oral intake being one factor among several rather than an automatic contraindication.

#418 · Airway Management, Procedures, and Ophthalmology

What is a key caution with propofol for procedural sedation?

Quick Reference

Propofol has no analgesic properties and can cause significant hypotension and respiratory depression. It is often combined with a short-acting opioid for painful procedures and requires close hemodynamic monitoring.

#419 · Airway Management, Procedures, and Ophthalmology

What reversal agents pair with opioid and benzodiazepine sedation?

Quick Reference

Naloxone reverses opioids and flumazenil reverses benzodiazepines. Flumazenil should be used cautiously in patients with chronic benzodiazepine use due to seizure risk from acute withdrawal.

#420 · Airway Management, Procedures, and Ophthalmology

How should escalating pain management be approached in the ED for severe pain?

Quick Reference

Start with the mildest effective option, such as acetaminophen or NSAIDs, and escalate to opioids for severe pain, but severe pain such as a long bone fracture should not be under-treated by rigidly stepping through a ladder; appropriate initial opioid dosing is acceptable for severe injuries.

#421 · Airway Management, Procedures, and Ophthalmology

Trap: NSAIDs are always a safe first-line analgesic choice.

Quick Reference

False. NSAIDs carry risk in renal impairment, active GI bleeding, and in some cardiovascular or pregnant patients, so analgesic choice must always be individualized to comorbidities.

#422 · Airway Management, Procedures, and Ophthalmology

What advantage does a regional nerve block offer over systemic opioids for an isolated limb injury?

Quick Reference

A nerve block, such as a hematoma or fascia iliaca block, can provide targeted, longer-lasting analgesia with fewer systemic side effects like sedation, nausea, or respiratory depression, compared to systemic opioids.

#423 · Airway Management, Procedures, and Ophthalmology

What risk factor requires extra caution in pediatric procedural sedation?

Quick Reference

Underlying airway abnormalities, obstructive sleep apnea, or a recent upper respiratory infection increase the risk of adverse respiratory events during sedation and should prompt a more cautious approach or deferral.

#424 · Airway Management, Procedures, and Ophthalmology

What is the use case for nitrous oxide in the ED?

Quick Reference

It is useful for brief, moderately painful procedures such as fracture reduction, laceration repair, or IV starts in cooperative patients, with rapid onset and offset and a good safety profile, though it provides only mild to moderate sedation and analgesia.

#425 · Airway Management, Procedures, and Ophthalmology

What are appropriate discharge criteria after procedural sedation?

Quick Reference

Return to baseline mental status, stable vital signs, adequate pain control, and ability to tolerate oral intake when relevant, along with a responsible adult for transport if any residual sedation risk remains.

#426 · Airway Management, Procedures, and Ophthalmology

What is the utility and drawback of etomidate for procedural sedation?

Quick Reference

Etomidate provides rapid onset with hemodynamic stability, useful for brief painful reductions, but it carries a notable risk of myoclonus and does not provide any analgesia on its own.

#427 · Airway Management, Procedures, and Ophthalmology

Trap: pain scores should be the sole determinant of opioid dosing regardless of clinical context.

Quick Reference

False. While self-reported pain scores matter, dosing should also account for the specific injury severity, vital sign trends, mental status, and response to initial doses, rather than letting a single number alone dictate treatment.

#428 · Airway Management, Procedures, and Ophthalmology

What is the classic presentation of acute angle-closure glaucoma?

Quick Reference

Sudden severe eye pain, blurred vision, halos around lights, headache, nausea, and a fixed mid-dilated pupil with a firm globe on palpation. It is a true ophthalmologic emergency requiring immediate pressure-lowering treatment.

#429 · Airway Management, Procedures, and Ophthalmology

What is the immediate ED treatment for suspected acute angle-closure glaucoma?

Quick Reference

Topical beta-blocker, alpha-agonist, and pilocarpine, plus a systemic carbonic anhydrase inhibitor such as acetazolamide if not contraindicated, with emergent ophthalmology consultation for definitive laser iridotomy.

#430 · Airway Management, Procedures, and Ophthalmology

What signs suggest an open globe injury and how should the exam be modified?

Quick Reference

An irregular pupil, extruded ocular contents, severe subconjunctival hemorrhage, or a positive Seidel sign suggest open globe injury. Avoid applying pressure, further manipulation, or tonometry, and protect the eye with a rigid shield instead.

#431 · Airway Management, Procedures, and Ophthalmology

What is the classic presentation of central retinal artery occlusion and why is timing critical?

Quick Reference

Sudden painless monocular vision loss, described as a curtain coming down, with a cherry-red spot on fundoscopy. The treatment window is very narrow and outcomes are poor if not addressed within a few hours, so emergent ophthalmology involvement should be arranged immediately.

#432 · Airway Management, Procedures, and Ophthalmology

What features distinguish temporal, or giant cell, arteritis presenting with vision loss?

Quick Reference

Age typically over 50, a new headache, jaw claudication, scalp tenderness, elevated ESR or CRP, and risk of sudden irreversible vision loss if untreated. High-dose corticosteroids should be started immediately on clinical suspicion, before biopsy confirmation.

#433 · Airway Management, Procedures, and Ophthalmology

Trap: wait for temporal artery biopsy results before starting steroids in suspected giant cell arteritis.

Quick Reference

False. Steroids should be started immediately based on clinical suspicion, since vision loss can become permanent within hours to days. Biopsy can still show pathology for up to 1 to 2 weeks after steroids are started.

#434 · Airway Management, Procedures, and Ophthalmology

What warning sign on exam predicts ocular involvement in herpes zoster ophthalmicus?

Quick Reference

Hutchinson's sign, a vesicular rash on the tip or side of the nose, predicts a higher likelihood of ocular involvement and should prompt urgent ophthalmology referral and antiviral treatment.

#435 · Airway Management, Procedures, and Ophthalmology

What distinguishes a corneal ulcer from a simple corneal abrasion?

Quick Reference

A corneal ulcer shows a white or gray infiltrate or opacity on the cornea along with the epithelial defect, suggesting infection, whereas a simple abrasion has a clear cornea with only an epithelial defect that takes up fluorescein.

#436 · Airway Management, Procedures, and Ophthalmology

What key diagnosis must be excluded in a contact lens wearer with a red, painful eye?

Quick Reference

Pseudomonas keratitis, a bacterial corneal ulcer, must be considered in any contact lens wearer with a red, painful eye, and patching is avoided since it can promote bacterial growth in this population.

#437 · Airway Management, Procedures, and Ophthalmology

What is the management principle for a traumatic hyphema?

Quick Reference

Elevate the head of the bed, protect the eye with a rigid shield, avoid NSAIDs and other agents that increase bleeding risk, and involve ophthalmology, since the risk of rebleeding and elevated intraocular pressure can threaten vision.

#438 · Airway Management, Procedures, and Ophthalmology

What is the immediate management priority for a chemical eye injury before any other history or exam?

Quick Reference

Immediate copious irrigation with saline or Ringer's lactate for at least 15 to 30 minutes should begin before detailed history, visual acuity testing, or waiting for pH paper, since delay worsens outcomes, especially with alkali burns which penetrate deeper than acid.

#439 · Airway Management, Procedures, and Ophthalmology

Why are alkali chemical burns to the eye generally worse than acid burns?

Quick Reference

Alkali substances cause liquefactive necrosis and progressively penetrate deeper into ocular tissue, while acids cause coagulative necrosis that tends to limit further penetration, making alkali injuries typically more severe and requiring more prolonged irrigation.

#440 · Airway Management, Procedures, and Ophthalmology

What features are classically associated with optic neuritis?

Quick Reference

Painful vision loss that is worse with eye movement, a relative afferent pupillary defect, and possible dyschromatopsia, or reduced color vision. It is strongly associated with multiple sclerosis and warrants MRI and neurology referral.

#441 · Airway Management, Procedures, and Ophthalmology

Trap: a normal-appearing eye on external exam excludes serious pathology in a patient with painless vision loss.

Quick Reference

False. Retinal artery occlusion, retinal detachment, and optic neuritis can all present with a completely normal-looking external eye exam, so visual acuity, pupillary reflexes, and fundoscopy or urgent ophthalmology exam are required even when the eye looks unremarkable.

#442 · Systems-Based Practice and EMS

Distinguish a medical directive from online medical control in EMS systems.

Quick Reference

A medical directive is a pre-approved standing protocol paramedics can follow without a real-time call, also called offline or indirect medical control, while online or direct medical control involves real-time physician communication for decisions outside standard protocols.

#443 · Systems-Based Practice and EMS

What is the base hospital physician's core role?

Quick Reference

Providing real-time direction to paramedics for situations outside standing orders, along with quality assurance review of paramedic performance and protocol compliance after calls.

#444 · Systems-Based Practice and EMS

What is the rationale for bypassing the closest hospital in field trauma triage?

Quick Reference

Patients meeting major trauma criteria should bypass a closer non-trauma center for a designated trauma center when the transport time difference is reasonable, since definitive trauma care improves outcomes more than proximity alone.

#445 · Systems-Based Practice and EMS

What is the general principle behind termination of resuscitation in the field?

Quick Reference

Paramedics may terminate resuscitation efforts in the field for adult non-traumatic cardiac arrest meeting specific validated criteria, such as an unwitnessed arrest, no bystander CPR, no shock advised, and no ROSC after full ALS care, guided by protocol or online medical control.

#446 · Systems-Based Practice and EMS

What is a physician's responsibility when directing EMS by radio or phone?

Quick Reference

Clearly identify yourself, confirm understanding of the paramedic's assessment, give explicit orders within the paramedic's scope of practice, and document the interaction, since the directing physician bears medico-legal responsibility for that directed care.

#447 · Systems-Based Practice and EMS

What is the scene safety principle for a physician who responds to a prehospital scene?

Quick Reference

Personal and scene safety always take priority before patient care, since an incapacitated responder helps no one and can worsen the overall incident.

#448 · Systems-Based Practice and EMS

What does the black triage tag color mean in mass casualty triage?

Quick Reference

Black indicates deceased or expectant, meaning injuries incompatible with survival given available resources. Resources are directed instead to patients more likely to benefit, a deliberate shift from usual individual patient-first care.

#449 · Systems-Based Practice and EMS

What is the first assessment step in the START triage algorithm?

Quick Reference

Ambulatory status. Patients who can walk are triaged as walking wounded, or green, first, quickly separating the majority of less critical patients so responders can focus on non-ambulatory patients needing respiratory, perfusion, and mental status assessment.

#450 · Systems-Based Practice and EMS

Trap: air medical helicopter transport is always faster and always improves outcomes.

Quick Reference

False. Helicopter transport carries its own risks and time costs such as activation, weather, and landing zone setup, and should be reserved for situations where the time savings and capability genuinely change outcome, not used reflexively.

#451 · Systems-Based Practice and EMS

What is the current practice shift in prehospital spinal motion restriction?

Quick Reference

Routine rigid backboard immobilization of all trauma patients has fallen out of favor due to complications such as pressure injuries and respiratory compromise with no proven benefit, in favor of selective spinal motion restriction based on clinical criteria.

#452 · Systems-Based Practice and EMS

What is a physician's role in local EMS quality improvement?

Quick Reference

Reviewing patient care reports against protocol adherence and outcomes, providing feedback to individual providers and the service, and identifying system-level gaps rather than focusing only on single-case errors.

#453 · Systems-Based Practice and EMS

What is the value of prehospital STEMI notification?

Quick Reference

Prehospital 12-lead ECG with advance notification to the receiving hospital or cath lab significantly reduces door-to-balloon time, since the team can mobilize before the patient physically arrives.

#454 · Systems-Based Practice and EMS

What is the general Good Samaritan legal protection principle for physicians in Canada?

Quick Reference

Physicians who provide emergency assistance in good faith outside their normal work setting are generally protected from liability for ordinary negligence, provided they act reasonably and within their competence, though this protection is not absolute.

#455 · Systems-Based Practice and EMS

Trap: a physician present at a prehospital scene automatically assumes full legal authority over EMS care.

Quick Reference

False. EMS providers operate under their own medical directives and base hospital physician oversight. A bystander physician can offer assistance and should identify their credentials, but formal transfer of authority and documentation matters, and paramedics may still need to confirm direction through their own system.

#456 · Systems-Based Practice and EMS

What is the ethical shift in disaster triage compared to usual clinical practice?

Quick Reference

Normal practice maximizes benefit to the individual patient in front of you. Disaster or mass casualty triage shifts to maximizing survivors across the whole population, which can mean withholding maximal resuscitation from an individual who would consume disproportionate resources.

#457 · Systems-Based Practice and EMS

Define surge capacity.

Quick Reference

The ability of a health system to expand rapidly beyond normal capacity in space, staff, supplies, and systems to manage a sudden increase in patient volume, such as a mass casualty incident or a pandemic surge.

#458 · Systems-Based Practice and EMS

What is the purpose of an incident command system in a hospital mass casualty response?

Quick Reference

A standardized management structure with clear roles such as incident commander, operations, logistics, and planning that ensures coordinated communication and resource allocation during a large-scale emergency, rather than ad hoc decision-making.

#459 · Systems-Based Practice and EMS

Explain the re-triage concept during an evolving mass casualty incident.

Quick Reference

Triage category is not fixed at first assessment. Patients must be reassessed periodically since condition can change, either deteriorating or stabilizing, and available resources may also ease or worsen as the incident evolves.

#460 · Systems-Based Practice and EMS

Trap: during a mass casualty incident, the sickest patient should always be treated first regardless of survivability.

Quick Reference

False. In true resource-limited mass casualty triage, an extremely unstable patient with a low likelihood of survival even with maximal intervention may be deprioritized in favor of salvageable patients who need less resource investment, a deliberate departure from usual one-patient-at-a-time ethics.

#461 · Systems-Based Practice and EMS

What communication principle helps when managing several critically ill patients simultaneously in the ED?

Quick Reference

Explicitly assign roles and use closed-loop communication with team members for each patient, and use a team leader who is not personally performing hands-on procedures so they can maintain overall situational awareness across all patients.

#462 · Systems-Based Practice and EMS

What is the priority for decontamination before treating a patient exposed to a hazardous chemical or radiologic agent?

Quick Reference

Decontamination must occur before the patient enters the main treatment area, to prevent secondary contamination of staff and equipment, even if this delays initial treatment slightly.

#463 · Systems-Based Practice and EMS

What consideration is distinct in a pandemic or infectious mass casualty event compared to a typical trauma mass casualty incident?

Quick Reference

Cohorting and isolation of infectious patients, appropriate personal protective equipment for staff, and altered crisis standards of care may apply, along with different resource bottlenecks such as ventilators and isolation beds than a typical trauma incident.

#464 · Systems-Based Practice and EMS

What is just-in-time training and why is it relevant during a sudden surge event?

Quick Reference

Brief, focused training given right before a task is needed, such as quickly teaching non-specialist staff a specific triage tool or procedure, to rapidly expand the effective workforce during an unexpected surge.

#465 · Systems-Based Practice and EMS

What documentation challenge is unique to mass casualty events and how is it mitigated?

Quick Reference

Standard full charting is often impossible in real time during a mass casualty event. Simplified standardized triage tags and abbreviated documentation systems are used to track patients without slowing care, with full documentation completed retrospectively once the acute phase passes.

#466 · Systems-Based Practice and EMS

What is the principle behind physician wellness and fatigue management during a prolonged multi-patient surge?

Quick Reference

Building in rest and rotation for staff during a prolonged incident preserves decision-making quality and patient safety, since fatigued providers make more errors. Sustainable staffing is itself a patient safety intervention, not a luxury.

#467 · Systems-Based Practice and EMS

Why is family reunification and communication planning relevant in a mass casualty response?

Quick Reference

A designated area and process for family inquiries and reunification reduces chaos in clinical areas and should be planned as part of the overall incident response, not handled ad hoc by clinical staff.

#468 · Systems-Based Practice and EMS

Trap: hospital mass casualty plans only need to address the initial surge of patients.

Quick Reference

False. Plans must also address secondary surge, such as delayed arrivals and transferred patients from overwhelmed facilities, ongoing operations like staffing and supply resupply, and eventual stand-down and recovery phases.

#469 · Systems-Based Practice and EMS

What is the key difference between prehospital field triage and ED arrival triage during a mass casualty event?

Quick Reference

Field triage is typically rapid and simplified, like START, to sort large numbers of patients quickly for transport, while ED triage upon arrival can reassess with more detailed information and resources, potentially changing a patient's triage category based on more thorough assessment.

#470 · Systems-Based Practice and EMS

Define continuous quality improvement in a clinical department context.

Quick Reference

A structured, ongoing process of measuring outcomes and processes of care, identifying gaps, implementing changes, and re-measuring to drive incremental improvement, rather than a one-time fix.

#471 · Systems-Based Practice and EMS

What is the purpose of the Plan-Do-Study-Act cycle?

Quick Reference

A rapid-cycle framework for testing a small change on a limited scale, planning the change, doing it, studying the results, and acting on what was learned, before committing to department-wide implementation.

#472 · Systems-Based Practice and EMS

What is the purpose of a root cause analysis after a serious adverse event?

Quick Reference

A structured, systems-focused review that asks why an error happened at every contributing level, not just what the individual did wrong, aiming to identify system fixes that prevent recurrence rather than assigning individual blame alone.

#473 · Systems-Based Practice and EMS

Trap: quality improvement initiatives should focus mainly on identifying which individual made the error.

Quick Reference

False. Modern quality improvement is grounded in systems thinking, recognizing most errors result from flawed processes, communication breakdowns, or system design rather than a single person's carelessness. Blaming individuals discourages honest error reporting.

#474 · Systems-Based Practice and EMS

Explain the Swiss cheese model of error.

Quick Reference

Multiple layers of defense, each with individual holes or weaknesses, must align for a serious error to reach a patient. Effective safety systems add and strengthen multiple independent layers rather than relying on any single safeguard.

#475 · Systems-Based Practice and EMS

Define structure, process, and outcome measures in the Donabedian model.

Quick Reference

Structure measures assess resources and systems in place, such as staffing ratios or equipment availability. Process measures assess whether recommended care steps were actually performed, such as door-to-needle time. Outcome measures assess the actual patient result, such as mortality or complication rate.

#476 · Systems-Based Practice and EMS

Why are process measures often used more than outcome measures for real-time quality monitoring?

Quick Reference

Outcome measures can be affected by many confounding factors and take longer to accumulate meaningful numbers, while process measures, like time to antibiotics in sepsis, are more immediately actionable and directly reflect the care delivered.

#477 · Systems-Based Practice and EMS

Define a sentinel event.

Quick Reference

An unexpected occurrence involving death or serious physical or psychological injury unrelated to the natural course of the patient's illness, which triggers a mandatory formal investigation given its severity.

#478 · Systems-Based Practice and EMS

Explain the just culture principle in patient safety.

Quick Reference

A framework that distinguishes between human error, at-risk behavior, and reckless behavior, applying supportive coaching for the first two and appropriate accountability only for genuinely reckless conduct, rather than punishing every error equally.

#479 · Systems-Based Practice and EMS

What role does closed-loop communication play in reducing medical error?

Quick Reference

Explicitly confirming that an instruction was heard and understood correctly, through a repeat-back, reduces communication-based errors, particularly during high-stress resuscitations with multiple simultaneous verbal orders.

#480 · Systems-Based Practice and EMS

Trap: a single successful PDSA test cycle is sufficient evidence to permanently change department-wide practice.

Quick Reference

False. PDSA is meant to be iterative. A small successful test should be refined and repeated at increasing scale before being adopted as standard practice department-wide.

#481 · Systems-Based Practice and EMS

What is the role of benchmarking in a departmental quality improvement program?

Quick Reference

Comparing a department's own performance metrics against similar departments or established standards helps identify whether a gap is a true local problem or already an accepted industry-wide limitation, guiding where improvement effort should be focused.

#482 · Systems-Based Practice and EMS

Give an example of a structured handover communication tool and explain its purpose.

Quick Reference

SBAR, standing for Situation, Background, Assessment, and Recommendation, standardizes handover communication between shifts or providers to reduce omission of critical information during transitions of care, a recognized high-risk point for error.

#483 · Systems-Based Practice and EMS

Why does reporting near misses, not just actual harm events, matter for a quality improvement program?

Quick Reference

Near misses reveal the same underlying system weaknesses as actual harm events but without the emotional and legal weight of patient injury, offering a much larger and less biased dataset to identify and fix systemic risks before they cause real harm.

#484 · Systems-Based Practice and EMS

What effect does a likelihood ratio of 0.1 have on post-test probability?

Quick Reference

An LR of 0.1 substantially lowers the probability of disease after a negative test result. Using a nomogram or Bayes' formula, even a moderate pretest probability drops to a low post-test probability, making this a strong rule-out test result.

#485 · Systems-Based Practice and EMS

What effect does a likelihood ratio of 10 have on post-test probability?

Quick Reference

An LR of 10 substantially raises the probability of disease after a positive test result, generally moving even a low-to-moderate pretest probability into a high post-test probability range, making this a strong rule-in finding.

#486 · Systems-Based Practice and EMS

Applied clinically, which matters more for ruling out disease, sensitivity or specificity?

Quick Reference

High sensitivity tests are best for ruling out disease when negative, sometimes remembered as SnNout, since few true cases are missed. High specificity tests are best for ruling in disease when positive, remembered as SpPin, since false positives are rare.

#487 · Systems-Based Practice and EMS

How should number needed to treat be interpreted for a clinical decision?

Quick Reference

NNT is the number of patients who need to receive a treatment for one additional patient to benefit compared to control. A lower NNT indicates a more impactful treatment effect, and it should be weighed against the number needed to harm before adopting a therapy.

#488 · Systems-Based Practice and EMS

Why does the distinction between absolute risk reduction and relative risk reduction matter clinically?

Quick Reference

Relative risk reduction can make a small absolute benefit look dramatic, since a 50 percent relative reduction from 2 percent to 1 percent risk is only a 1 percent absolute reduction. Absolute risk reduction gives a more clinically meaningful picture of true benefit for an individual patient.

#489 · Systems-Based Practice and EMS

What is the confidence interval interpretation trap to watch for?

Quick Reference

A 95 percent confidence interval that crosses the line of no effect, meaning 1.0 for a ratio measure or 0 for a difference measure, means the result is not statistically significant, even if the point estimate itself looks clinically impressive.

#490 · Systems-Based Practice and EMS

Explain the difference between statistical significance and clinical significance.

Quick Reference

A result can be statistically significant, unlikely due to chance and often driven by a large sample size, without being clinically meaningful in magnitude. Conversely, a clinically important effect can fail to reach statistical significance in an underpowered study, so both must be considered together.

#491 · Systems-Based Practice and EMS

What is the main methodologic advantage of a randomized controlled trial over an observational study for causal inference?

Quick Reference

Randomization balances both known and unknown confounding variables between groups, allowing more confident causal conclusions about a treatment's effect than observational designs, which remain vulnerable to confounding no matter how well they adjust statistically.

#492 · Systems-Based Practice and EMS

Define publication bias and explain why it matters for reading the literature.

Quick Reference

The tendency for studies with positive or statistically significant results to be published more often than negative or null studies, which can make a treatment or test look more effective in the aggregate published literature than it truly is.

#493 · Systems-Based Practice and EMS

What is the purpose of an intention-to-treat analysis?

Quick Reference

Analyzing patients according to their originally assigned group regardless of whether they actually completed or adhered to the assigned treatment, preserving the benefit of randomization and giving a more realistic estimate of real-world effectiveness.

#494 · Systems-Based Practice and EMS

Give an example of selection bias relevant to emergency medicine research.

Quick Reference

A study recruiting patients only during daytime weekday hours may systematically miss sicker or different patients who present overnight or on weekends, skewing results and limiting how well the findings generalize to the full ED population.

#495 · Systems-Based Practice and EMS

Why is pretest probability central to interpreting any diagnostic test result?

Quick Reference

The same test result, whether positive or negative, produces a very different post-test probability depending on the pretest probability the clinician started with. A test cannot be interpreted correctly in isolation from the specific patient's prior likelihood of disease.

#496 · Systems-Based Practice and EMS

Trap: a study with a large sample size and a very low p-value automatically means the finding is clinically important and free of bias.

Quick Reference

False. A low p-value only addresses the likelihood the result is due to chance, not whether the study was well-designed, free of confounding and bias, or whether the effect size is large enough to actually matter for patient care.